COMPUTATIONAL METHODS

(ES-201)

UNIT-I

Rolle's Theorem:-

If a function $f(x)$ is such that

(i) It is continuous in the closed interval $[a,b]$

(ii) It is derivable in the open interval (a,b)

(iii) $f(a) = f(b)$ then, there exists at least one value 'c' of x in the open interval (a,b) such that $f'(c) = 0$.

Geometrical proof: Let AB be the graph of the function $y = f(x)$ such that the point A and B of the graph correspond to the numbers a and b of the interval $[a,b]$.

1. As $f(x)$ is continuous in the interval $a \leq x \leq b$, therefore its graph is continuous curve from A to B.
2. As $f(x)$ is derivable in the interval $a < x < b$, therefore its graph has a unique tangent at every point between A and B.
3. As $(a) = f(b)$.

Therefore, AL and BM, the ordinates of A and B are equal.

It is evident from the figures that there is at least one point P on the curve between A and B, the tangent at which is parallel to x-axis.

Therefore, tangent at P is parallel to x-axis.

Therefore, slope of tangent at P is zero.

If 'c' is the abscissa of P, then $f'(c) = 0$ where $a < c < b$.

Note:-

1. Rolle's Theorem fails to hold good for a function which does not satisfy even one of the three conditions stated above.
2. Every polynomial is a continuous function of x for every value of x. $Sinx$, $Cosx$, e^x are continuous for all values of x. $logx$ is continuous for all $x > 0$.
3. Alternate form of Rolle's Theorem. If a function $f(x)$ is such that
   (i) It is continuous in the interval $a \leq x \leq a + h$
   (ii) It is derivable in the interval $a < x < a + h$
   (iii) $f(a) = f(a + h)$

Then there exists at least one number θ such that $f'(a + \theta h) = 0,\ 0 < \theta < 1$.

Because the number c which lies between a and $a + h$ must be greater than a by a fraction of h and may be written as $c = a + \theta h$ where $0 < \theta < 1$.

Example1. Verify Rolle's Theorem for the function $f(x) = x^2 - 6x + 8$ in the interval [2, 4].

Sol. Here $a = 2, b = 4$

1. $f(x)$ is a polynomial, since every polynomial is a continuous function of x for every value of x.

   Therefore, $f(x)$ is continuous in {2, 4].

2. $f'^{(x)} = 2x - 6$ which exist in the open interval (2, 4).
3. $f(2) = 4 - 12 + 8 = 0$ and $f(4) = 16 - 24 + 8 = 0$

   Therefore,

$$f(2) = 0 = f(4)$$

Therefore, $f(x)$ satisfies all the three conditions of Rolle's Theorem.

Therefore, there must exist at least one number c between 2 and 4 such that $f'(c) = 0$

Now $f'(x) = 2x - 6$

Therefore,

$f'(c) = 2c - 6 = 0$ so $c = 3$.

This is a point in the given open interval (2, 4) and, therefore, the Theorem is verified.

Example2. Verify Rolle's Theorem for $f(x) = (x-a)^m(x-b)^n$ in the interval $[a,b]$; where , m, n being positive integers.

Sol. $c = \frac{mb+na}{m+n}$

Example3. Verify Rolle's Theorem for $f(x) = 2 + (x-1)^{\frac{2}{3}}$ in the interval $[0,2]$.

Sol. Given function is

$f(x) = 2 + (x-1)^{\frac{2}{3}}$

Therefore,

$f'(x) = \frac{2}{3(x-1)^{\frac{1}{3}}}$

Which does not exist at $x = 1 \in (0,2)$

Example4. Verify Rolle's Theorem for $f(x) = \frac{sinx}{e^x}$ in $[0, \pi]$.

Ans. Roll's theorem exist and $c = \frac{\pi}{4}$

Example5. $f(x) = tanx$ in $[0, \pi]$.

Ans. Not exist because $f(x) = tanx$ is not continuous at $x = \frac{\pi}{2}$

Example6. $f(x) = |x|$ in $[-1,1]$.

Ans. Not exist because $f(x) = |x|$ is not derivable at $x = 0$.

Lagrange's Mean Value Theorem:-

If a function $f(x)$ is such that

(i) It is continuous in the closed interval $[a,b]$

(ii) It is derivable in the open interval (a,b)

Then there exists at least one value $'c'$ of x in the open interval (a,b) such that

$f'(c) = \frac{f(b)-f(a)}{b-a}$.

Proof. Consider the function

$F(x) = f(x) + Ax$ (1)

Where A is a constant to be determined such that $f(a) = F(b)$.

Now

$f(a) = f(a) + Aa$ and $f(b) = f(b) + Ab$

Since $f(a) = F(b)$

Therefore, $f(a) + Aa = f(b) + Ab$

Implies that $f(b) - f(a) = -A(b-a)$

Therefore, $-A = \frac{f(b)-f(a)}{b-a}$ (2)

Because $b - a \neq 0$ length of the interval

Now $f(x)$ is given to be continuous in $a \leq x \leq b$ and derivable in $a < x < b$.

Also, A being a constant, Ax is also continuous in $a \leq x \leq b$ and derivable in $a < x < b$.

Therefore,

$F(x) = f(x) + Ax$ is

1. Continuous in the interval $a \leq x \leq b$
2. Derivable in the interval $a < x < b$
3. $f(a) = F(b)$

Therefore, $F(x)$ satisfies all the three conditions of Rolle's Theorem.

Therefore, there must exist at least one value 'c' of x in the open interval (a, b) such that $f'(c) = 0$

Now $F'(x) = f'(x) + A$

Therefore, $F'(c) = f'(c) + A = 0$, implies that

$-A = f'(c)$ (3)

From (2) and (3), we have

$\frac{f(b)-f(a)}{b-a} = f'(c)$

Geometrical interpretation of Lagrange's Mean Value Theorem

Let A and B be points on the graph of the function $y = f(x)$ corresponding to $x = a$ and $x = b$.

Therefore, the coordinates of the points A and B are $\big(a, f(a)\big)$ and $\big(b, f(b)\big)$ respectively.

Slope of the chord AB $= \frac{Difference\ of\ ordinates}{Difference\ of\ abscissa} = \frac{f(b)-f(a)}{b-a}$

Now

1. Since $f(x)$ is continuous in $a \leq x \leq b$.

Therefore, its graph is continuous curve from A to B.

2. Since $f(x)$ is derivable in $a < x < b$, therefore, it possesses a unique tangent at every point between A and B.

It is evident from the figures that there is at least one point P between A and B, the tangent at which is parallel to chord AB.

If c is the abscissa of this point, then slope of tangent there at is $f'(c)$

Hence $\frac{f(b)-f(a)}{b-a} = f'(c)$.

Example1. Verify mean value theorem for $f(x) = x(x-1)(x-2)$ in $\left[0, \frac{1}{2}\right]$.

Solution. Since $f(x)$ being a polynomial is continuous in $\left[0, \frac{1}{2}\right]$

Also $f'(x) = 3x^2 - 6x + 2$ which exists in interval $\left(0, \frac{1}{2}\right)$

Therefore, by mean value theorem, we have

$$f'(c) = \frac{f\left(\frac{1}{2}\right) - f(0)}{\frac{1}{2} - 0}$$

$$3c^2 - 6c + 2 = \frac{\left(\frac{1}{8} - \frac{3}{4} + 1\right) - (0)}{\frac{1}{2} - 0} = \frac{3}{4}$$

$12c^2 - 24c + 5 = 0$, implies that

$c = 1.76, 0.24$

Discarding the value $c = 1.76$ which does not lie in the given interval $\left(0, \frac{1}{2}\right)$

Therefore, $= 0.24$, a value which lies in the interval $\left(0, \frac{1}{2}\right)$.

Hence mean value theorem verified.

Example2. Verify mean value theorem for the following functions

(a) $f(x) = \sqrt{x^2 - 4}$ in $[2, 4]$ Ans. $c = \sqrt{6}$

(b) $f(x) = logx$ in $[1, e]$ Ans. $c = e - 1$

(c) $f(x) = \frac{2x-1}{3x-4}$in$[1, 2]$ Ans. $f(x)$ is not continuous at $x = \frac{4}{3}$.

Example3. Show that if $x > 0, \log(1 + x) > \frac{x}{1+x}$

Solition. Let $\mathrm{f}(\mathrm{x}) = \log(1 + x) - \frac{x}{1+x}$

Therefore,

$f'(x) = \frac{x}{(1+x)^2}$ which is positive, because $x > 0$

Therefore, $\mathrm{f}(\mathrm{x})$ is an increasing function when $x > 0$

Implies that

$\mathrm{f}(\mathrm{x}) > f(0)$

Now, $\mathrm{f}(0) = \log 1 - 0 = 0$

Therefore,

$\mathrm{f}(\mathrm{x}) > 0$ implies that $\log(1 + x) - \frac{x}{1+x} > 0.$

Implies that

$\log(1 + x) > \frac{x}{1+x}$

Cauchy's Mean Value Theorem

If two functions $f(x)$ and $\varphi(x)$ are such that

1. Both are continuous in the closed interval $[a,b]$
2. Both are derivable in the open interval (a,b)
3. $\varphi'(x) \neq 0$ for any value of x in the open interval (a,b) then there exists at least one value c of x in the open interval (a,b) such that
   $\frac{f(b)-f(a)}{\varphi(b)-\varphi(a)} = \frac{f'(c)}{\varphi'(c)}$

Proof. Consider the function $F(x) = f(x) + A\varphi(x)$ (1)

Where A is a constant to be determined such that $F(a) = F(b)$

Now

$F(a) = f(a) + A\varphi(a)$ and $F(b) = f(b) + A\varphi(b)$

Therefore,

$f(a) + A\varphi(a) = f(b) + A\varphi(b)$, implies that

$-A = \frac{f(b)-f(a)}{\varphi(b)-\varphi(a)}$ where $\varphi(b) - \varphi(a) \neq 0$ (2)

If $\varphi(b) - \varphi(a) = 0,$ then $\varphi(b) = \varphi(a)$

So $\varphi(x)$ satisfies all the three conditions of Rolle 's Theorem.

Therefore, $\varphi'(x) = 0$ for at least one value x in the open interval $a < x < b$ which is contrary to the given condition that $\varphi'(x) \neq 0$ for any value of x in the open interval $a < x < b$.

Since $f(x)$ and $\varphi(x)$ are both given to be continuous in the interval $a \leq x \leq b$ and derivable in the open interval $a < x < b$.

Therefore, $F(x) = f(x) + A\varphi(x)$ is

1. continuous in the closed interval $[a,b]$
2. derivable in the open interval (a,b)
3. $F(a) = F(b)$

Therefore, $F(x)$ satisfies all the three conditions of Rolle's Theorem. So there exists at least one value c of x in the open interval (a,b) such that $F'(c) = 0$

Now

$F'(x) = f'(x) + A\varphi'(x)$

Therefore,

$F'(c) = 0$ gives $f'(c) + A\varphi'(c) = 0$ implies that

$$-A = \frac{f'(c)}{\varphi'(c)} \qquad (3)$$

From (2) and (3) we have

$$\frac{f(b)-f(a)}{\varphi(b)-\varphi(a)} = \frac{f'(c)}{\varphi'(c)}$$

Example1. Find $'c'$ of Cauchy's Mean Value Theorem for the following pairs of functions in $[a,b]$:

Where $f(x) = e^x$, $\varphi(x) = e^{-x}$.

Solution. Since $f(x) = e^x$, $\varphi(x) = e^{-x}$ are both continuous in $[a,b]$ and both functions are differential in (a,b).

Therefore,

$f'(x) = e^x$, $\varphi'(x) = -e^{-x}$

By Cauchy's mean value theorem

$$\frac{f(b)-f(a}{g(b)-g(a)} = \frac{f'(c)}{g'(c)}$$

$\frac{e^b-e^a}{e^{-b}-e^{-a}} = \frac{e^c}{e^{-c}}$ implies that

$$c = \frac{1}{2}(a+b)$$

Thus c lies in (a,b) which verifies the Cauchy's Mean Value theorem.

Taylor's Theorem

If a function $f(x)$ such that

1. $f(x)$, $f'(x)$, $f''(x)$, ……,$f^{n-1}(x)$are continuous in the closed interval $a \le x \le a+h$.
2. $f^n(x)$ exist in the open interval $a < x < a+h$, then there exist one number θ between 0 and 1 such that

$$f(a+h) = f(a) + hf'(a) + \frac{h^2}{2!}f''(a) + \frac{h^3}{3!}f'''(a) + \ldots.. + \frac{h^{(n-1)}}{(n-1)!}f^{(n-1)}(a) + \quad \frac{h^n}{n!}f^n(a+\theta h)$$

The $(n+1)^{th}$ term $\frac{h^n}{n!}f^n(a+\theta h)$ is called Lagrange's form of remainder after n

Terms and is denoted by R_n

Note:

Alternate form of Taylor's theorem with Lagrange's form of remainder after n terms.

If we put $a+h=b$, then the interval $[a,a+h]$ becomes $[a,b]$

Since $a+\theta h$ is a number between a and $a+h$ i.e. between $a$and b, let $a+\theta h=c$ where $a<c<b$.

Then the above theorem becomes

$$f(b)=f(a)+(b-a)f'(a)+\frac{(b-a)^2}{2!}f''(a)+\frac{(b-a)^3}{3!}f'''(a)+\ .....$$

$$...+\frac{(b-a)^{(n-1)}}{(n-1)!}f^{(n-1)}(a)+\quad\frac{(b-a)^n}{n!}f^n(c)$$

Where $a<c<b$.

Taylor's series

1. $f(x+h)=f(x)+hf'(x)+\frac{h^2}{2!}f''(x)+\frac{h^3}{3!}f'''(x)+\ .....$
2. $f(x+h)=f(h)+xf'(h)+\frac{x^2}{2!}f''(h)+\frac{x^3}{3!}f'''(h)+\ .....$
3. To $f(x)$ in ascending integral powers of $(x-a)$, replacing x by a and h by $(x-a)$ in (1), we get

   $f(x)=f[a+(x-a)]=f(a)+(x-a)f'(a)+\frac{(x-a)^2}{2!}f''(a)+\frac{(x-a)^3}{3!}f'''(a)+\ .....+$

   $\frac{(x-a)^n}{n!}f^n(a)+\ .....$

Example1. Use Taylor's series to prove that

$$logsin(x+h)=logsinx+hcotx-\frac{h^2}{2}cosec^2x+\frac{h^3}{3}cotxcosec^2x+...$$

Solution. Let $f(x+h)=logsin(x+h)$

Putting $h=0$, $f(x)=logsinx$

$f'(x) = cotx$, $f''(x) = -cosec^2x$, $f'''(x) = 2cosec^2xcotx$, ...

$f(x+h) = f(x) + hf'(x) + \frac{(h)^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + ...$

$logsin(x+h) = logsinx + hcotx - \frac{h^2}{2}cosec^2x + \frac{h^3}{3}cotxcosec^2x + ...$

Example 2. If $f(x) = x^3 + 2x^2 - 5x + 11$, calculate the value of $f\left(\frac{9}{10}\right)$ using Taylor's series.

Solution. To evaluate $f\left(\frac{9}{10}\right) = f\left(1 - \frac{1}{10}\right)$ therefore, $x = 1$ and $h = -\frac{1}{10}$

By Taylor's series

$f(x+h) = f(x) + hf'(x) + \frac{(h)^2}{2!}f''(x) + \frac{h^3}{3!}f'''(x) + ...$ (1)

Putting $x = 1$ and $h = -\frac{1}{10}$ in (1), we get

$f\left(\frac{9}{10}\right) = f(1) + \left(-\frac{1}{10}\right)f'(1) + \frac{\left(-\frac{1}{10}\right)^2}{2!}f''(x) + \frac{\left(-\frac{1}{10}\right)^3}{3!}f'''(1) + ...$ (2)

Here, $f(x) = x^3 + 2x^2 - 5x + 11$ therefore, $f(1) = 9$

$f'(x) = 3x^2 + 4x - 5 f'(1) = 2$

$f''(x) = 6x + 4 f''(1) = 10$

$f'''(x) = 6 f'''(1) = 6$

All higher derivatives becomes zero, therefore, from (2), we get

$f\left(\frac{9}{10}\right) = 9 - \frac{1}{10}.2 + \frac{1}{2}.\frac{1}{100}.10 - \frac{1}{6}.\frac{1}{1000}.6 = 8.849$

Example3. Using Taylor's series express the polynomial $2x^3 + 7x^2 + x - 6$ in powers of $(x-2)$.

Solution. Let $f(x) = 2x^3 + 7x^2 + x - 6$ (1)

Now we write $(x) = f[2 + (x-2)]$,

Therefore, $x = 2$ and $h = x - 2$

By Taylor's series

$$f(x) = f[2 + (x-2)] = f(2) + hf'(2) + \frac{(h)^2}{2!}f''(2) + \frac{(h)^3}{3!}f'''(2) \qquad (2)$$

Here, from (1)

$$f(x) = 2x^3 + 7x^2 + x - 6$$

Therefore, $f(2) = 40$

$$f'(x) = 6x^2 + 14x + 1 f'(2) = 53$$

$$f''(x) = 12x + 14 f''(2) = 38$$

$$f'''(x) = 12 f'''(2) = 12$$

All higher derivatives becomes zero, therefore, from (2), we get

$$2x^3 + 7x^2 + x - 6 = 40 + (x-2)53 + \frac{(x-2)^2}{2!}38 + \frac{(x-2)^3}{3!}12$$

OR

$$2x^3 + 7x^2 + x - 6 = 40 + 53(x-2) + 19(x-2)^2 + 2(x-2)^3$$

Approximation and Errors

The input information is rarely exact since it comes from some measurement or the other and the method also introduces further error. As such, the error in the final result may be due to error in the initial data or in the method or both. Our effort will be to minimize these errors, so as to get best possible results. We therefore begin by explaining various kinds of approximations and errors which may occur in a problem and derive some results on error propagation in numerical calculations.

Accuracy of numbers

1. **Approximate numbers**: There are two types of numbers exact and approximate. Exact numbers are 2, 4, 9, 13, 7/2, 6.45, ...etc. But there are numbers such as 4/3 (= 1.3333333.....), $\sqrt{2}$ (= 1.414213.....) and $\pi(= 3.141592.....)$ which cannot be expressed by a finite number of digits. These may be approximated by numbers 1.3333, 1.4142 and 3.1416 respectively. Such numbers which represent the given numbers to a certain degree of accuracy are called approximate numbers.
2. **Significant figures:** The digits used to express a number are called significant digits (figures). Thus each of the numbers 7845, 3.589, 0.4758 contains four significant figures, while the numbers 0.00386, 0.000587 and 0.0000296 contains only three significant figures since zeros only help to fix the position of the decimal point. Similarly the numbers 45000 and 7300.00 have two significant figures only.
3. **Rounding off:** There are numbers with large number of digits e,g, 22/7=3.142857143. In practice, it is desirable to limit such numbers to a manageable number of digits such as 3.14 or 3.143. This process of dropping unwanted digits is called rounding off.

   Rule : Rule to round off a number to n significant figures:

   (i) Discard all digits to the right of the nth.

   (ii) If this discard number is

   (a) Less than half a unit in the nth place, leave the nth digit unchanged.

   (b) Greater than half a unit in the nth place, increase the nth digit by unity.

(c) Exactly half a unit in the nth place, increase the nth digit by unity if it is odd otherwise leave it unchanged.

For instance, the following numbers rounded off to three significant figures are :

7.893	7.89
3.567	3.57
12.865	12.9
84767	84800
6.4356	6.44
5.8254	5.82

Also the following numbers rounded off to four places of decimal are:

6.284359	6.2844
9.864651	9.8646
12.464762	12.4648

Note- The numbers thus rounded off to n significant figures (or n decimal places) are said to be correct to n significant figures (or n decimal places).

Errors

In any numerical computation, we come across the following types of errors:

1. **Inherent errors:-** Errors which are already present in the statement of a problem before its solution, are called inherent errors. Such errors arise either due to the given data being approximate or due to the limitations of mathematical tables, calculators or the digital computer. Inherent errors can be minimized by taking better data or by using high precision computing aids.
2. **Rounding errors:**- Arise from the process of rounding off the numbers during the computation. Such errors are unavoidable in most of the calculations due to the limitations of the computing

aids. Rounding errors can, however, be reduced by retaining at least one more significant figure at each step than that given in the data and rounding off at the last step.

3. **Truncation errors:**- truncation errors are caused by using approximate results or on replacing an infinite process by a finite one. If we are using a decimal computer having a fixed word length of 4 digits, rounding off of 13.658 gives 13.66 whereas truncation gives 13.65.
   For example, if $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!} + ...\infty = X$ (say). Is replaced by $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} = X'$ (say), then the truncation error is $X - X'$. Truncation error is a type of algorithm error.
4. **Absolute, Relative and Percentage errors:**- If X is the true value of a quantity and X' is its approximate value, then $|X - X'|$ $i.e.$ $|error|$ is called the absolute error denoted as E_a.
   The absolute error is $E_a = |X - X'|$
   The relative error is defined by $E_r = \left|\frac{X-X'}{X}\right|$ $i.e.$ $\left|\frac{Error}{True\ Error}\right|$
   The percentage error is $E_p = 100E_r = 100\left|\frac{X-X'}{X}\right|$
   If $\overline{X}$ be such a number that $\left|X - \overline{X}\right| \leq \overline{X}$, then $\overline{X}$ is an upper limit on the magnitude of absolute error and measures the absolute accuracy.

Useful Rules for Estimating Errors

To estimate the errors which creep in when the numbers in a calculation are truncated or rounded off to a certain number of digits, the following rules are useful.

If the approximate value of a number X having n decimal digits is X', then

1. Absolute error due to truncation to $k$digits = $|X - X'| < 10^{n-k}$.
2. Absolute error due to rounding off to $k$digits = $|X - X'| < \frac{1}{2}.10^{n-k}$.
3. Relative error due to truncation to $k$digits = $\left|\frac{X-X'}{X}\right| < 10^{1-k}$.
4. Relative error due to rounding off to$k$ digits = $\left|\frac{X-X'}{X}\right| < \frac{1}{2}.10^{1-k}$.

Example1. Round off the numbers 865250 and 37.46235 to four significant figures and compute $E_a, E_r E_p$ in each case.

Solution. (i) Number rounded off to four significant figures = 865200

Therefore,

$E_a = |X - X'| = |865250 - 865200| = 50$

$E_r = \left|\frac{X-X'}{X}\right| = \frac{50}{865250} = 6.71 \times 10^{-5}$

$E_p = 100E_r = 6.27 \times 10^{-3}$

(ii) Number rounded off to four significant figures = 37.46

$E_a = |X - X'| = |37.46235 - 37.46| = 0.00235$

$E_r = \left|\frac{X-X'}{X}\right| = \frac{0.00235}{37.46235} = 6.27 \times 10^{-5}$

$E_p = 100E_r = 6.27 \times 10^{-3}.$

Example2. Find the absolute error if the number $X = 0.00545828$ is

(i) Truncated to three decimal digits.

(ii) Rounded off to three decimal digits.

Solution. We have $X = 0.00545828 = 0.545828 \times 10^{-2}$

(i) After truncating to three decimal places,

its approximate value

$X' = 0.545828 \times 10^{-2}$ Therefore,

Absolute error

$E_a = |X - X'| = |0.00545828 - 0.00546| = 0.000828 \times 10^{-2} = 0.828 \times 10^{-5} < 10^{-5}$

(ii) After truncating to three decimal places,

its approximate value

$X' = 0.546 \times 10^{-2}$

Therefore,

Absolute error

$E_a = \left|X - X'\right| = |0.00545828 - 0.00546|$

$= 0.000172 \times 10^{-2} = 0.172 \times 10^{-5} < 10^{-5}$

Example3. Find the relative error if the number $X = 0.004997$ is

(i) Truncated to three decimal digits.

(ii) Rounded off to three decimal digits.

Solution. We have $X = 0.004997 = 0.4997 \times 10^{-2}$

(i) After truncated to three decimal places,

its approximate value

$X' = 0.499 \times 10^{-2}$

Therefore,

Relative error $E_r = \left|\frac{X - X'}{X}\right| = \left|\frac{0.4997 \times 10^{-2} - 0.499 \times 10^{-2}}{0.4997 \times 10^{-2}}\right|$

$= 0.140 \times 10^{-2} < \times 10^{1-3}$

(ii) After rounding off to three decimal places,

its approximate value

$X' = 0.500 \times 10^{-2}$

Therefore,

Relative error $E_r = \left|\frac{X - X'}{X}\right| = \left|\frac{0.4997 \times 10^{-2} - 0.500 \times 10^{-2}}{0.4997 \times 10^{-2}}\right|$

$= 0.600 \times 10^{-3} = 0.06 \times 10^{-3+1} < 0.5 \times 10^{1-3}$

LOCATION OF ROOTS FOR ALGEBRIC AND TRANSCENDENTAL EQUATIONS

Introduction

In Engineering Mathematics we often encounter problems of obtaining solutions of equations of the form $f(x) = 0$. In other words we have to find a number x_0 such that $f(x_0) = 0$.

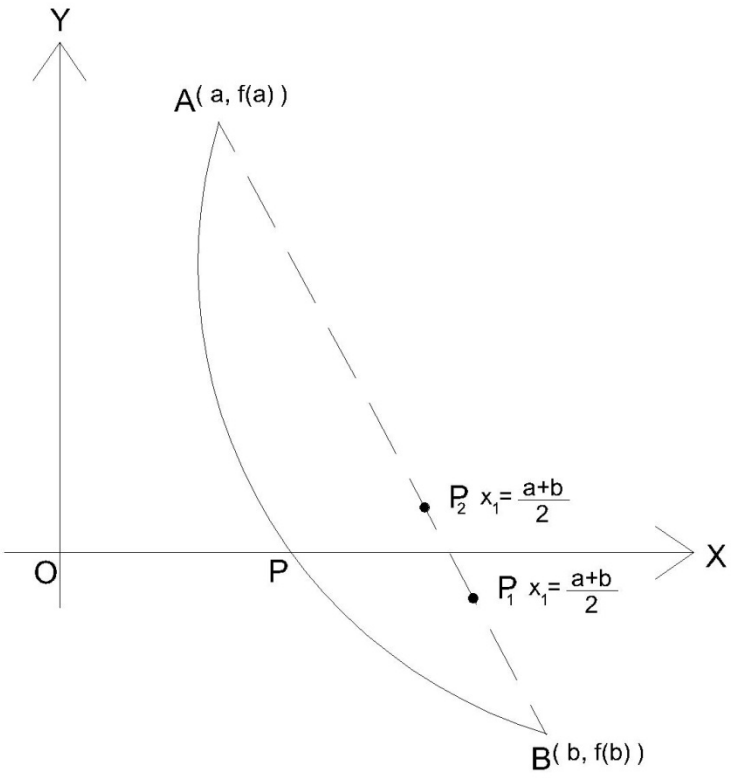
As we know to solve a quadratic equation $f(x) = ax^2 + bx + c = 0$, we have a simple formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$ to find its roots. If $f(x)$ is a polynomial of higher degree or an expression involving transcendental functions such as circular, logarithmic or exponential functions there is no direct method for solving these equations, so such equations are solved by numerical methods. In these methods, to locate the root of an equation $f(x) = 0$ we use the following theorem in calculus.

If $f(x)$ is continuous in the interval $[a,b]$ and if $f(a)$ and $f(b)$ are of opposite sign then the equation $f(x) = 0$ has at least one root lying between $a$and b.

First we find an interval in which the root lies. If α and β are two numbers such that $f(\alpha)$ and $f(\beta)$ have opposite signs, then a root of $f(x) = 0$ lies between α and β. Here, we discussing following methods to obtain approximate solutions for algebraic and transcendental equations.

Bisection method (Bolzano method):

Let $f(x)$ be a continuous function between a and b such that $f(a)$ and $f(b)$ are of opposite signs. Then there is a root of $f(x) = 0$ lies between $a$and b. Assuming that $f(a) > 0$and $f(b) < 0$. By bisection method, let first approximation root is $x_1 = \frac{a+b}{2}$.



If $f\left(x_1\right) = 0$, then x_1 is a root of $f\left(x\right) = 0$. If $f(x) \neq 0$, then root lies between a and x_1 or x_1 and b depending on whether $f\left(x_1\right) > 0$ or $f\left(x_1\right) < 0$. Further bisect the interval in which the root lies and continue the process until theroot is found to the desired accuracy.

Note:- The bisection is simple but the convergence is very slow. The interval within which the root lies is bisected each time until we get the root with desired accuracy.

Rate of Convergence:- The fastness of convergence in any method is known as rate of convergence.

For Bisection method:-

Since the new interval containing the root, is exactly half the length of the previous one, the interval width is reduced by a factor of ½ at each step. At the end of the nth step, the new interval will therefore, be of length $\frac{(b-a)}{2^n}$. If on repeating this process n times, the latest interval is as small as givenϵ,

then

$\frac{(b-a)}{2^n} \leq \epsilon$ (1)

Taking log on both sides of (1), we get

$log\,(b-a) - n\log\,(2) \leq \log\,(\epsilon)$

$n\log(2) \geq \log(b-a) - \log\,(\epsilon)$

$n \geq \frac{[\log(b-a) - log\epsilon]}{log2}$

This gives the number of iterations required for achieving accuracyϵ,

As the error decreases with each step by a factor of ½, $\left(i.e.\ \frac{e_{n+1}}{e_n} = \frac{1}{2}\right)$

The convergence in the bisection method is linear.

Example1. Find real root of the equation $x^3 - x - 11 = 0$ by using bisection method.

Solution: Let $f(x) = x^3 - x - 11$, then $f(2) = -5$ and $f(3) = 13$, therefore one root lies between 2 and 3.

Let $a = 2$ and $b = 3$.

The first approximation $x_1 = \frac{2+3}{2} = 2.5$

Now $f(2.5) = 2.125$ which is positive and $f(2)$ is negative.

Therefore root lies between 2 and 2.5

Now let $a = 2$and $b = 2.5$, therefore second approximation $x_2 = \frac{2+2.5}{2} = 2.25$

Now $f(2.25) = -1.8594$ and therefore root lies between 2.25 and 2.5.

This process is repeated and the calculation of the successive approximations is given as below:

	a	b	$x_i = \frac{a+b}{2}$	$f(x_i)$
1	2	3	$x_1 = 2.5$	2.125
2	2	2.5	$x_2 = 2.25$	− 1.18594
3	2.25	2.5	$x_3 = 2.375$	0.0215
4	2.25	2.375	$x_4 = 2.3125$	− 0.946
5	2.3125	2.375	$x_5 = 2.3438$	− 0.4684
6	2.3438	2.375	$x_6 = 2.3594$	− 0.2252

And so on

Therefore root is 2.3 up to one decimal place.

Example2. Find positive real root of $x\log_{10}x = 1.2$ using the bisection method in three iterations.

Solution: Let $f(x) = x\log_{10}x - 1.2$,

$f(2) = 2\log_{10}2 - 1.2 = -0.598 = -ve$;

$f(3) = 3\log_{10}3 - 1.2 = 0.2313 = +ve$

Hence root lies between 2 and 3. Let $a = 2$and$b = 3$.

The first approximation is $x_1 = \frac{a+b}{2} = \frac{2+3}{2} = 2.5$.

Now $f(2.5) = 2.5\log_{10}(2.5) - 1.2 = 2.5 \times 0.3979 - 1.2 = -0.2053 = -ve$

Since $f(2.5) < 0$ and $f(3) > 0$ the root lies between 2.5 and 3.

Let $a = 2.5$ and $b = 3$.

Therefore, second approximation is $x_2 = \frac{a+b}{2} = \frac{2.5+3}{2} = 2.75$.

Now $f(2.75) = 2.75\log_{10}(2.75) - 1.2 = 2.75 \times 0.4393 - 1.2 = 0.008 = +ve$.

Since $f(2.75) > 0$ and $f(2.5) < 0$ the root lies between 2.5 and 2.75.

Let $a = 2.5$and $b = 2.75$.

Therefore, third approximation is

$x_3 = \frac{a+b}{2} = \frac{2.5+2.75}{2} = 2.625$.

Now $f(2.625) = 2.625\log_{10}(2.625) - 1.2 = 2.625 \times 0.4191 - 1.2 = -0.10 = -ve$.

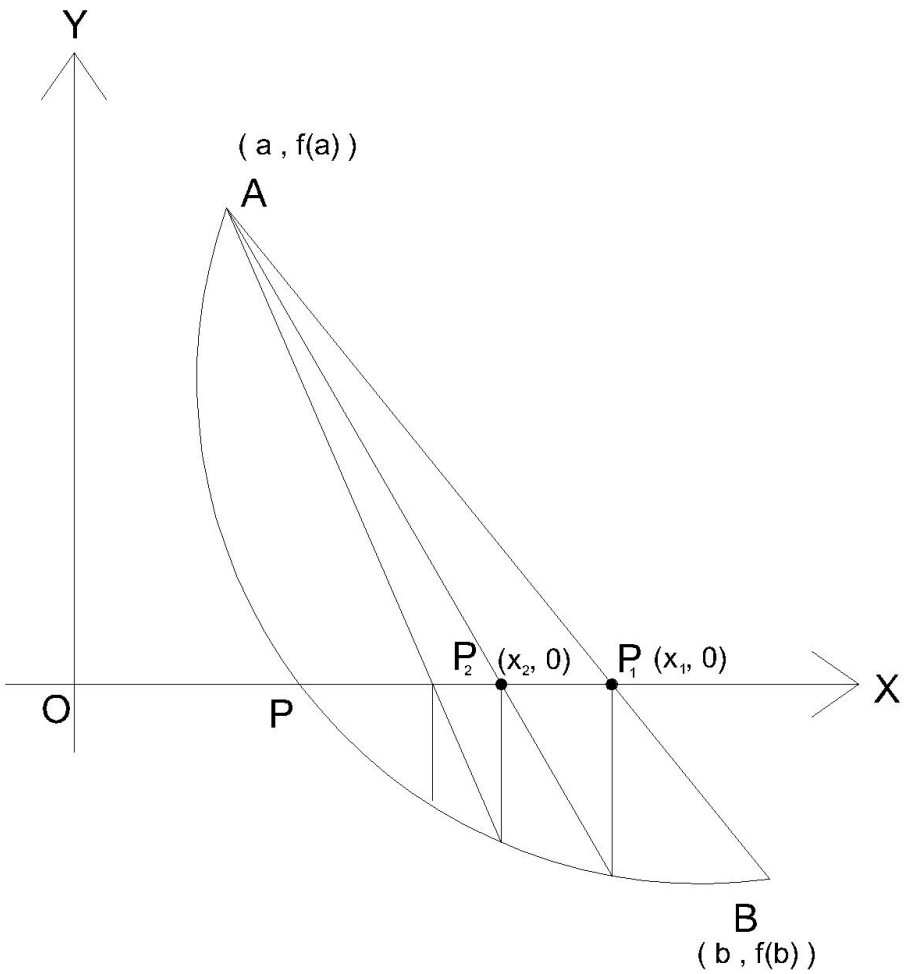
Since $f(2.625) < 0$ and $f(2.75) > 0$ the root lies between 2.625 and 2.75 and so on.

RegulaFalsi method (Method of false position)

Consider the continuous function $y = f(x)$. Taking two points a, b such that $f(a)$ and $f(b)$ are of opposite signs, therefore one root lies between $a$and b. Let graph of the function $y = f(x)$ passes through two points $A\left(a,f(a)\right)$ and $B\left(b,f(b)\right)$ and this graph of $y = f(x)$ crosses the $x-$ axis between these points, This indicates that a root lies between $a\ and\ b$consquently $f(a).f(b) < 0$.

Equation of the chord joining the points A and B is given as

$$y - f(a) = \frac{f(b)-f(a)}{b-a}(x-a) \qquad (1)$$



When this chord meets the x-axis at point $\left(x_1,0\right)$ where x_1 is the first approximation to the root. Putting $y = 0$ in (1), we get

$0 - f(a) = \frac{f(b)-f(a)}{b-a}(x_1 - a)$, implies that

$x_1 = a - \frac{b-a}{f(b)-f(a)} f(a)$

Hence $x_1 = \frac{af(b)-bf(a)}{f(b)-f(a)}$ (2)

Which is an approximation to the root.

Now we find $f(x_1)$, if $f(a)$ and $f(x_1)$ are of opposite signs then the root lies between a and x_1. So we replace b by x_1 in (2) and get the next approximation x_2.

But if $f(a)$ and $f(x_1)$ are of same sign, then $f(x_1)$ and $f(b)$ will be of opposite signs. Hence root lies between $x_1$and b, then we replace a by x_1 in (2).

Repeated this process until the root is found to the desired accuracy.

Example1. Find a root of the equation $x^3 - 3x - 5 = 0$ by the method of false position up to two places of decimal.

Solution: Let $f(x) = x^3 - 3x - 5$, now $f(1) = -1$ and $f(2) = 3$,

Therefore, one root lies between 1 and 2.

Let $a = 1$and $b = 2$, therefore the first approximation is

$x_1 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{1f(2)-2f(1)}{f(2)-f(1)} = \frac{1\times3-2\times(-1)}{2-(-1)} = 1.25$

Now, $f(x_1) = f(1.25) = -0.7969$. Since $f(1.25)$ is –ve and $f(2)$ is +ve ,

Therefore, root lies between 1.25 and 2.

Let $a = 1.25$and $b = 2$. The second approximation is

$x_2 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{1.25f(2)-2f(1.25)}{f(2)-f(1.25)} = \frac{1.25\times3-2\times(-0.7969)}{3-(-0.7969)} = 1.4074.$

Now $f(1.4074) = (1.4074)^3 - 3 \times 1.4074 + 1 = -0.4345.$

Since $f(1.4074)$ is negative and $f(2)$ is positive,

Therefore, the root lies between 1.4074 and 2.

Taking $a = 1.4074$and $b = 2$,

Therefore, the third approximation is

$$x_3 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{1.4074f(2)-2f(1.4074)}{f(2)-f(1.4074)} = \frac{1.4074\times3-2\times(-0.4345)}{3-(-0.4345)} = 1.4824.$$

Now $f(1.4824) = (1.4824)^3 - 3\times1.4824 + 1 = -0.1896.$

Since $f(1.4824)$ is negative and $f(2)$ is positive,

Therefore, the root lies between 1.4824 and 2.

Taking $a = 1.4824$and $b = 2$, therefore, the fourth approximation is

$$x_4 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{1.4824f(2)-2f(1.4824)}{f(2)-f(1.4824)} = \frac{1.4824\times3-2\times(-0.1896)}{3-(-0.1896)} = 1.5132.$$

Now $f(1.5132) = (1.5132)^3 - 3\times1.5132 + 1 = -0.0747.$

Since $f(1.5132)$ is negative and $f(2)$ is positive,

Therefore, the root lies between 1.5132 and 2.

Taking $a = 1.5132$and $b = 2$,

Therefore, the fifth approximation is

$$x_5 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{1.5132f(2)-2f(1.5132)}{f(2)-f(1.5132)} = \frac{1.5132\times3-2\times(-0.0747)}{3-(-0.0747)} = 1.525.$$

Now $f(1.525) = (1.525)^3 - 3\times1.525 + 1 = -0.0284.$

Since $f(1.525)$ is negative and $f(2)$ is positive,

Therefore, the root lies between 1.525 and 2.

Taking $a = 1.525$and $b = 2$, therefore, the sixth approximation is

$$x_6 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{1.525f(2)-2f(1.525)}{f(2)-f(1.525)} = \frac{1.5132\times3-2\times(-0.0284)}{3-(-0.0284)} = 1.5295.$$

Therefore, $x_5 = x_6 = 1.52$So required root is 1.52 (corrected to 3 places of decimal).

Example2. Find the smallest positive root of $x^2 - log_e x - 12 = 0$ by Regulafalsi method.

Solution: Let $f(x) = x^2 - log_e x - 12,$

$f(3) = 9 - 1.0986 - 12 = -4.0986$; $f(4) = 16 - 1.3863 - 12 = 2.6137$

Therefore, root lies between 3 and 4. Let $a = 3$ and $b = 4$

The first approximation to the root is

$x_1 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{3f(4)-4f(3)}{f(4)-f(3)} = \frac{3\times 2.6137-4\times(-4.0986)}{2.6137-(-4.0986)} = 3.6106.$

Now $f\left(x_1\right) = f\left(3.6106\right) = (3.6106)^2 - 1.2839 - 12 = -0.2474.$

Since $f\left(3.6106\right)$ is negative and $f\left(4\right)$ is positive,

Therefore, the root lies between 3.6106 and 4.

Taking $a = 3.6106$and $b = 4$,

Therefore, the second approximation is

$x_2 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{3.6106\times 2.6137-4\times(-0.2474)}{2.6137-(-0.2474)} = 3.6443.$

Now $f\left(x_2\right) = f\left(3.6443\right) = (3.6443)^2 - 1.2932 - 12 = -0.0123.$

Since $f\left(3.6443\right)$ is negative and $f\left(4\right)$ is positive,

Therefore, the root lies between 3.6443 and 4.

Taking $a = 3.6443$and $b = 4$,

Therefore, the third approximation is

$x_3 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{3.6443\times 2.6137-4\times(-0.0123)}{2.6137-(-0.0123)} = 3.646.$

Now $f\left(x_3\right) = f\left(3.646\right) = (3.646)^2 - 1.2936 - 12 = -0.0003.$

Since $f\left(3.646\right)$ is negative and $f\left(4\right)$ is positive

Therefore, the root lies between 3.646 and 4.

Taking $a = 3.646$and $b = 4$,

Therefore, the fourth approximation is

$x_4 = \frac{af(b)-bf(a)}{f(b)-f(a)} = \frac{3.646\times 2.6137-4\times(-0.0003)}{2.6137-(-0.0003)} = 3.6461.$

Hence the required root is 3.646 (corrected up to 3 decimals).

Secant Method:-

This method is an improvement over the method of false position as it does not require the condition $f(x_0)\,f(x_1) < 0$ of that method. Here also the graph of the function $y = f(x)$ is approximated by a secant line but at each iteration, two most recent approximations to the root are used to find the next approximation. Also it is not necessary that the interval must contain the root.

Taking x_0 and x_1 as the initial limits of the interval, we write the equation of the chord joining these as

$$y - f(x_1) = \frac{f(x_1)-f(x_0)}{x_1-x_0}(x - x_1)$$

Then the abscissa of the point where it crosses the x-axis (y=0) is given by

$$x_2 = x_1 - \frac{x_1-x_0}{f(x_1)-f(x_0)}f(x_1)$$

Which is an approximation to the root.

The general formula for successive approximations is, therefore, given by

$$x_{n+1} = x_n - \frac{x_n-x_{n-1}}{f(x_n)-f(x_{n-1})}f(x_n),\ n \geq 1$$

Note:- If at any iteration $f(x_n) = f(x_{n-1})$, this method fails and shows that it does not converge. This is a drawback of secant method over the method of false position which always converges. But if the secant method once converges, its rate of convergence is 1.6 which is faster than that of the method of false position.

Example1. Find a root of the equation $x^3 - 2x - 5 = 0$ using secant method correct to three decimal places.

Solution. Let $f(x) = x^3 - 2x - 5$ so that

$f(2) = -1$ and $f(3) = 16$

Taking initial approximations $x_0 = 2$ and $x_1 = 3$, by secant method, we have

$$x_2 = x_1 - \frac{x_1-x_0}{f(x_1)-f(x_0)}f(x_1) = 3 - \frac{3-2}{16+1}16 = 2.058823$$

Now $f(x_2) = -0.390799$

Therefore,

$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2) = 2.058823 - \left(\frac{2.058823 - 3}{-0.390799 - 16}\right) \times (-0.390799)$

$= 2.081263$

Now $f(x_3) = -0.147204.$

Therefore,

$x_4 = x_3 - \frac{x_3 - x_2}{f(x_3) - f(x_2)} f(x_3)$

$= 2.081263 - \left(\frac{2.081263 - 2.058823}{-0.147204 - (-0.390799)}\right) \times (-0.147204)$

$= 2.094824$

Now $f(x_4) = 0.003042$

$x_5 = x_4 - \frac{x_4 - x_3}{f(x_4) - f(x_3)} f(x_4)$

$= 2.094824 - \left(\frac{2.094824 - 2.081263}{0003042 - (-0.147204)}\right) \times (0.003042)$

$= 2.094549$

Hence the root is 2.094 correct to three decimal places.

Example2. Find the root of the equation $xe^x = cosx$ using secant method correct to three decimal places.

Solution. Let $f(x) = xe^x - cosx = 0$

$f(0) = 0 - 1 = -1 = -ve$

$f(1) = 1.e^1 - \cos(1) = 3.14 - 0.5403 = 2.5997 = +ve$

Therefore, root of the given equation lies within $(0,1)$.

Taking initial approximations $x_0 = 0$ and $x_1 = 1$, by secant method, we have

$x_2 = x_1 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_1)$

$x_2 = 1 - \left(\frac{1-0}{0.5403-(-1)}\right) \times (0.5403) = 1 - 0.350776 = 0.649224$

$f(x_2) = 0.649224e^{0.649224} - \cos(0.649224)$

$= 0.649224 \times 1.91405 - 0.79655 = 0.44609$

$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2) = 0.649224 - \left(\frac{0.649224-1}{0.44609-2.5997}\right) \times 0.44609$

$= 0.649224—0.350776 = 0.649224—0.$

Newton-Raphson method (Method of tangents)

Let x_0 be an approximate root of the equation $f(x) = 0$. Let $x_1 = x_0 + h$ be the exact root where h very small, positive or negative.

Therefore, $f(x_1) = 0$ (1)

By Taylor's series expansion, we have

$f(x_1) = f(x_0 + h) = f(x_0) + \frac{h}{1!} f'(x_0) + \frac{h^2}{2!} f''(x_0) + \ldots$

Since $f(x_1) = 0$ and h is very small h^2 and higher powers of h can be neglected. Therefore

$f(x_0) + \frac{h}{1!} f'(x_0) = 0$, implies that

$h = -\frac{f(x_0)}{f'(x_0)}$ if $f'(x_0) \neq 0$. Hence

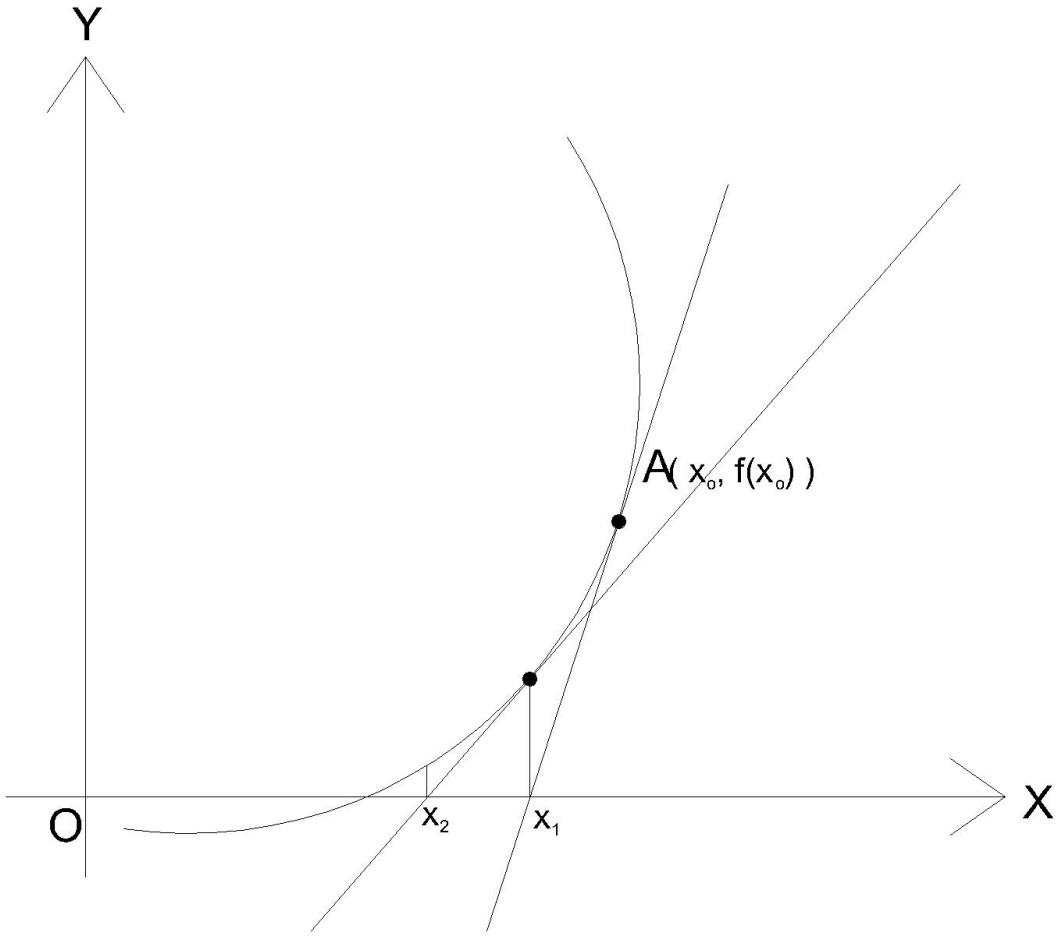
$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$ is a first approximation to the root.

Similarly starting with $x_1$we get the next approximation to the root given by

$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ and so on.

In general

$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, $n = 0,1,2,\ldots$ and it is known as Newton-Raphson iteration formula OR simply Newton-Raphson's formula.



Example1. Find the real root of $xe^x - 2 = 0$ correct to three places of decimals using Newton-Raphson method.

Solution: Let $f(x) = xe^x - 2$

Therefore, $f'(x) = xe^x + e^x = e^x(x+1)$.

Now $f(0) = -2$ and $f(1) = e - 2 = 2.7183 - 2 = 0.7183$.

Therefore root lies between 0 and 1.

Since numerical value of $f(1)$ is less than that of $f(0)$ we can take the initial approximation as $x_0 = 1$.

The Newton-Raphson formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}\ , n = 0,1,2,...$$

Therefore $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

where $f(x_0) = f(1) = 0.7183$;

$f'(x_0) = f'(1) = 2.7183(1+1) = 5.4366.$

Therefore,

$x_1 = 1 - \frac{0.7183}{5.4366} = 0.8679.$

Now, $f(x_1) = f(0.8679) = 0.0673$;

$f'(x_1) = f'(0.8679) = 4.4492.$

Therefore, $x_2 = x_1 - \frac{f(x_1)}{f'(1)} = 0.8679 - \frac{0.0673}{4.4492} = 0.8528.$

Now, $f(x_2) = f(0.8528) = 0.8528 \times e^{0.8528} - 2 = 0.0008$;

$f'(x_2) = f'(0.8528) = 4.3471.$

Therefore, $x_3 = x_2 - \frac{f(x_2)}{f'2} = 0.8528 - \frac{0.0008}{4.3471} = 0.8526.$

Since x_2 and x_3 are approximately equal, up to third decimals we can take the required root as 0.853.

Example2. Evaluate $\sqrt{12}$ to four places of decimals by Newton-Raphson method.

Solution: Let $x = \sqrt{12}$

Therefore, $x^2 - 12 = 0$.

Let $f(x) = x^2 - 12$. $f'(x) = 2x$.

Now $f(3) = -3$ and $f(4) = 4$.

Therefore root lies between 3 and 4.

Since numerically value of $f(3)$ is less than that of $f(4)$ we take the initial approximation as $x_0 = 3$.

The Newton-Raphson formula is

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \; n = 0,1,2,\ldots,$$

Therefore, $x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$

Now $f(x_0) = f(3) = 9 - 12 = -3$;

$f'(x_0) = f'(3) = 6$,

Therefore, $x_1 = 3 - \frac{-3}{6} = 3.5$.

Now $f(x_1) = f(3.5) = 12.25 - 12 = 0.25$;

$f'(x_1) = f'(3.5) = 7$, therefore

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)} = 3.5 - \frac{0.25}{7} = 3.4643.$$

Now $f(x_2) = f(3.4643) = 12.0014 - 12 = 0.0014$;

$f'(x_1) = f'(3.4643) = 6.9286$, therefore

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)} = 3.4643 - \frac{0.0014}{6.9286} = 3.4641.$$

The required root is 3.464 to three places of decimals.

Example3. Find the real root of $xe^x - 2 = 0$ correct to three places of decimals.

Solution: Let $f(x) = xe^x - 2$

Therefore, $f^{|}(x) = xe^x + e^x = e^x(x+1)$

We know that $f(0) = -2 = -ve$

$f(1) = e - 2 = 2.7183 - 2 = 0.7183 = +ve$

Therefore, root lies between $0 \text{ and } 1$

Since the numerical value of $f(1)$ is less than that of $f(0)$.

We can take the initial approximation as $x_0 = 1$.

The Newton-Raphson formula is

$x_{n+1} = x_n - \frac{f(x_n)}{f^{|}(x_n)}$;$n = 0, 1, 2, 3, \ldots$

Therefore, $x_1 = x_0 - \frac{f(x_0)}{f^{|}(x_0)}$

Now, $f(x_0) = f(1) = 0.7183$; $f^{|}(x_0) = f^{|}(1) = 2.7183(1+1) = 5.4366$

Therefore, $x_1 = x_0 - \frac{f(x_0)}{f^{|}(x_0)} = 1 - \frac{0.7183}{5.4366} = 1 - 0.1321 = 0.8679$

Now, $f(x_1) = f(0.8679) = 0.0673$; $f^{|}(x_1) = f^{|}(0.8679) = 4.4492$

Therefore, $x_2 = x_1 - \frac{f(x_1)}{f^{|}(x_1)} = 0.8679 - \frac{0.0673}{4.4492} = 0.8679 - 0.0151 = 0.8528$

Now, $f(x_2) = f(0.8528) = 0.8528e^{0.8528} - 2 = 0.0008$

$f^{|}(x_2) = f^{|}(0.8528) = e^{0.8528}(0.8528+1) = 4.3471$

$x_3 = x_2 - \frac{f(x_2)}{f^{|}(x_2)} = 0.8528 - \frac{0.0008}{4.3471} = 0.8526$

Since $x_2 \text{and} x_3$ are approximately equal, up to third decimals we can take the required root as 0.853.

Secant Method:-

This method is an improvement over the method of false position as it does not require the condition $f(x_0)\,f(x_1) < 0$ of that method. Here also the graph of the function $y = f(x)$ is approximated by a secant line but at each iteration, two most recent approximations to the root are used to find the next approximation. Also it is not necessary that the interval must contain the root.

Taking x_0 and x_1 as the initial limits of the interval, we write the equation of the chord joining these as

$$y - f(x_1) = \frac{f(x_1)-f(x_0)}{x_1-x_0}(x - x_1)$$

Then the abscissa of the point where it crosses the x-axis (y=0) is given by

$$x_2 = x_1 - \frac{x_1-x_0}{f(x_1)-f(x_0)}f(x_1)$$

Which is an approximation to the root. The general formula for successive approximations is, therefore, given by

$$x_{n+1} = x_n - \frac{x_n-x_{n-1}}{f(x_n)-f(x_{n-1})}f(x_n),\ n \geq 1$$

Note:- If at any iteration $f(x_n) = f(x_{n-1})$, this method fails and shows that it does not converge. This is a drawback of secant method over the method of false position which always converges. But if the secant method once converges, its rate of convergence is 1.6 which is faster than that of the method of false position.

Example1. Find a root of the equation $x^3 - 2x - 5 = 0$ using secant method correct to three decimal places.

Solution. Let $f(x) = x^3 - 2x - 5$ so that

$f(2) = -1$ and $f(3) = 16$

Taking initial approximations $x_0 = 2$ and $x_1 = 3$, by secant method, we have

$$x_2 = x_1 - \frac{x_1-x_0}{f(x_1)-f(x_0)}f(x_1) = 3 - \frac{3-2}{16+1}16 = 2.058823$$

Now $f(x_2) = -0.390799$

Therefore,

$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2) = 2.058823 - \left(\frac{2.058823 - 3}{-0.390799 - 16}\right) \times (-0.390799)$

$= 2.081263$

Now $f(x_3) = -0.147204.$

Therefore,

$x_4 = x_3 - \frac{x_3 - x_2}{f(x_3) - f(x_2)} f(x_3)$

$= 2.081263 - \left(\frac{2.081263 - 2.058823}{-0.147204 - (-0.390799)}\right) \times (-0.147204)$

$= 2.094824$

Now $f(x_4) = 0.003042$

$x_5 = x_4 - \frac{x_4 - x_3}{f(x_4) - f(x_3)} f(x_4)$

$= 2.094824 - \left(\frac{2.094824 - 2.081263}{0003042 - (-0.147204)}\right) \times (0.003042)$

$= 2.094549$

Hence the root is 2.094 correct to three decimal places.

Example2. Find the root of the equation $xe^x = cosx$ using secant method correct to three decimal places.

Solution. Let $f(x) = xe^x - cosx = 0$

$f(0) = 0 - 1 = -1 = -ve$

$f(1) = 1.e^1 - cos1 = 2.71828 - 0.540302 = 2.17797 = +ve$

Taking initial approximations $x_0 = 0$ and $x_1 = 1$, by secant method, we have

$x_2 = x_1 - \frac{x_1 - x_0}{f(x_1) - f(x_0)} f(x_1)$

$x_2 = 1 - \left(\frac{1 - 0}{2.17797 - (-1)}\right) (2.17797 = 0.31466$

$f(x_2) = f(0.31466) = 0.31466e^{0.31466} - \cos(0.31466)$

$= 0.431019 - 0.950901 = -0.519882$

$x_3 = x_2 - \frac{x_2 - x_1}{f(x_2) - f(x_1)} f(x_2)$

$x_3 = 0.31466 - \frac{0.31466 - 1}{-0.519882 - 2.17797} \times (-0.519882)$

$x_3 = 0.31466 — \left(\frac{-0.68543}{-2.697852}\right)(-0.519882) = 0.446743$

$f(x_3) = f(0.446743) = 0.446743e^{0.446743} - \cos(0.446743)$

$f(x_3) = 0.698354 - 0.901859 = -0.203505$

$x_4 = x_3 - \frac{x_3 - x_2}{f(x_3) - f(x_2)} f(x_3) = 0.446743 - \left(\frac{0.446743 - 0.31466}{-0.203505 - (-0.519882)}\right) \times (-0.203505)$

$x_4 = 0.446743 - \left(\frac{0.132083}{0.316377}\right) \times (-0.203505) = 0.531703$

Rate of Convergence

The fastness of convergence of a root of an equation is known as rate of convergence.

If ϵ_n and ϵ_{n+1} are the errors at nth and (n+1)th iterations, then

$\epsilon_{n+1} \propto (\epsilon_n)^k$ where k is known as order of convergence.

Rate of convergence of Scent Method:-

Let $f(x)$, $f^{!}(x)\ and\ f^{!!}(x)$ be all continuous functions in the neighborhood of the root $x=\alpha$ of the equation $f(x)=0$ and let $f^{!}(\alpha)\neq 0$. We will now show that if initially $x_1\ and\ x_2$ are chosen sufficiently close to the root $x=\alpha$, then the sequence $< x_n >$ converges to $x=\alpha$ and the order of convergence is 1.62 approximately.

Let $x_n=\alpha+\epsilon_n$, then $x_{n+1}=\alpha+\epsilon_{n+1}\ and\ x_{n-1}=\alpha+\epsilon_{n-1}$ (1)

We know that

$$x_{n+1}=\frac{x_n f(x_{n-1})-x_{n-1}f(x_n)}{f(x_{n-1})-f(x_n)}$$

$$\alpha+\epsilon_{n+1}=\frac{(\alpha+\epsilon_n)f(\alpha+\epsilon_{n-1})-(\alpha+\epsilon_{n-1})f(\alpha+\epsilon_n)}{f(\alpha+\epsilon_{n-1})-f(\alpha+\epsilon_n)}$$

$$\alpha+\epsilon_{n+1}=\frac{\alpha[f(\alpha+\epsilon_{n-1})-f(\alpha+\epsilon_n)]+\epsilon_n f(\alpha+\epsilon_{n-1})-\epsilon_{n-1}f(\alpha+\epsilon_n)}{f(\alpha+\epsilon_{n-1})-f(\alpha+\epsilon_n)}$$

$$\alpha+\epsilon_{n+1}=\alpha+\frac{\epsilon_n f(\alpha+\epsilon_{n-1})-\epsilon_{n-1}f(\alpha+\epsilon_n)}{f(\alpha+\epsilon_{n-1})-f(\alpha+\epsilon_n)}$$

$$\epsilon_{n+1}=\frac{\left[\epsilon_n\left\{f(\alpha)+\epsilon_{n-1}f^{!}(\alpha)+\frac{1}{2}\epsilon_{n-1}^2 f^{!!}(\alpha)+\ldots\right\}-\epsilon_{n-1}\left\{f(\alpha)+\epsilon_n f^{!}(\alpha)+\frac{1}{2}\epsilon_n^2 f^{!!}(\alpha)+\ldots\right\}\right]}{\left\{f(\alpha)+\epsilon_{n-1}f^{!}(\alpha)+\frac{1}{2}\epsilon_{n-1}^2 f^{!!}(\alpha)+\ldots\right\}-\left\{f(\alpha)+\epsilon_n f^{!}(\alpha)+\frac{1}{2}\epsilon_n^2 f^{!!}(\alpha)+\ldots\right\}}$$

$$\epsilon_{n+1}=\frac{f(\alpha)\{\epsilon_n-\epsilon_{n-1}\}+f^{!}(\alpha)\{\epsilon_n\epsilon_{n-1}-\epsilon_n\epsilon_{n-1}\}+\frac{1}{2}f^{!!}(\alpha)\epsilon_n\epsilon_{n-1}\{\epsilon_{n-1}-\epsilon_n\}+\ldots}{f^{!}(\alpha)\{\epsilon_n-\epsilon_{n-1}\}+\frac{1}{2}f^{!!}(\alpha)\left\{(\epsilon_{n-1})^2-(\epsilon_n)^2\right\}+\ldots}$$

Since α is the root

Therefore, $f(\alpha)=0$, we get

$$\epsilon_{n+1}=\frac{1}{2}\epsilon_n\varepsilon_{n-1}\frac{f^{!!}(\alpha)}{f^{!}(\alpha)}+O\left(\varepsilon_n^2\right),\ \text{as}\ f(a)=0,\ f^{!}(a)\neq 0$$

$$\epsilon_{n+1}=M\epsilon_n.\epsilon_{n-1}\ \text{where}\ M=\frac{1}{2}\frac{f^{!!}(\alpha)}{f^{!}(\alpha)} \quad (2)$$

Now we want to determine a number p, such that

$\epsilon_{n+1}=c\epsilon_n^p$, implies that $\epsilon_n=c\epsilon_{n-1}^p$ and hence

$$\epsilon_{n-1}=c^{-\frac{1}{p}}(\epsilon_n)^{\frac{1}{p}}$$

From (2), we get

$c\epsilon_n^p = M\epsilon_n c^{-\frac{1}{p}}(\epsilon_n)^{\frac{1}{p}}$

$c\epsilon_n^p = Mc^{-\frac{1}{p}}(\epsilon_n)^{1+\frac{1}{p}}$

Equating powers of ϵ on both sides, we get

$p = 1 + \frac{1}{p}$, implies that

$p^2 - p - 1 = 0$, implies that

$p = \frac{1\pm\sqrt{1+4}}{2} = \frac{1\pm\sqrt{5}}{2}$

Taking only positive value, so

$p = \frac{1+\sqrt{5}}{2} = 1.62$

Therefore,

$\epsilon_{n+1} = c\epsilon_n^p = c\epsilon_n^{1.62}$

$\epsilon_{n+1} \propto (\epsilon_n)^{1.62}$

Therefore, secant method rate of convergence of order 1.62 approximately.

Rate of convergence of Newton-Raphson method

Let x_n differ from the root α by a small quantity ϵ_n, so that

$x_n = \alpha + \epsilon_n$ and $x_{n+1} = \alpha + \epsilon_{n+1}$

Therefore, $x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$, we get

$\alpha + \epsilon_{n+1} = \alpha + \epsilon_n - \frac{f(\alpha+\epsilon_n)}{f'(\alpha+\epsilon_{n+1})}$

$\epsilon_{n+1} = \epsilon_n - \frac{f(\alpha+\epsilon_n)}{f'(\alpha+\epsilon_n)}$

$\epsilon_{n+1} = \epsilon_n - \frac{f(\alpha)+\epsilon_n f'(\alpha)+\frac{1}{2}(\epsilon_n)^2 f''(\alpha)+\ldots}{f'(\alpha)+\epsilon_n f''(\alpha)+\ldots}$

$$\epsilon_{n+1} = \frac{\epsilon_n f^{/}(\alpha) + (\epsilon_n)^2 f^{//}(\alpha) + \ldots - f(\alpha) - \epsilon_n f^{/}(\alpha) - \frac{1}{2}(\epsilon_n)^2 f^{//}(\alpha) - \ldots}{f^{/}(\alpha) + \epsilon_n f^{//}(\alpha) + \ldots}$$

Since α is the root

Therefore, $f(\alpha) = 0$ and neglecting higher powers of ϵ_n, we get

$$\epsilon_{n+1} = \frac{1}{2}(\epsilon_n)^2 \frac{f^{//}(\alpha)}{f^{/}(\alpha)}$$

$\epsilon_{n+1} = M(\epsilon_n)^2$ where $M = \frac{1}{2}\frac{f^{//}(\alpha)}{f^{/}(\alpha)}$ is a constant.

$\epsilon_{n+1} \propto (\epsilon_n)^2$

This shows that subsequent error at each step is proportional to the square of the previous error.

Therefore, Newton-Raphson method has **quadratic** rate of convergence.

Rate of convergence of bisection method

Let $f(x) = 0$ be an equation and suppose α is its root.

Therefore, $f(\alpha) = 0$

Let ϵ be the error in the root, so

$\epsilon_n = x_n - \alpha$, $\epsilon_{n+1} = x_{n+1} - \alpha$ *and* $\epsilon_{n-1} = x_{n-1} - \alpha$

Therefore,

$x_n = \epsilon_n + \alpha$, $x_{n+1} = \epsilon_{n+1} + \alpha$ *and* $x_{n-1} = \epsilon_{n-1} + \alpha$

We know that by bisection method,

$x_{n+1} = \frac{x_{n-1} + x_n}{2}$, implies that

$$\epsilon_{n+1} + \alpha = \frac{\epsilon_{n-1} + \alpha + \epsilon_n + \alpha}{2} = \frac{2\alpha + \epsilon_{n-1} + \epsilon_n}{2} = \alpha + \frac{\epsilon_{n-1} + \epsilon_n}{2}$$

$$\epsilon_{n+1} = \frac{\epsilon_{n-1} + \epsilon_n}{2} = \frac{1}{2}\epsilon_n \left(1 + \frac{\epsilon_{n-1}}{\epsilon_n}\right)$$

Neglecting $\frac{\epsilon_{n-1}}{\epsilon_n}$, which is very small.

Therefore,

$\epsilon_{n+1} = \frac{1}{2}\epsilon_n$ OR $\epsilon_{n+1} \propto (\epsilon_n)^1$

Therefore, rate of convergence of bisection method is linear.

NOTE:- For any method, k is known as order of convergence or rate of convergence in the relation

$\epsilon_{n+1} \propto (\epsilon_n)^k$

Where ϵ_n and ϵ_{n+1} represents errors in two consecutive iterations.

FIBONACCI SEARCH METHOD

The Fibonacci numbers $\{F_n\}$ are defined as

$F_0 = 1, F_1 = 1$ Then $F_n = F_{n-1} + F_{n-2}$, $n \geq 2$

Fibonacci sequence:- 1 , 1 , 2 , 3 , 5 , 8 , 13 , 21 , ...

$F_0, F_1, F_2, F_3, F_4, F_5$.....

Task is to minimize $\{F_n\}$ over $[a,b]$

Divide the interval $[a,b]$ equally into F_n subintervals and hence the length of each subinterval is $\frac{b-a}{F_n}$

$a - - ! - - ! - - ! - - ! - - ! - - ! - - ! - - ! - - ! - - ! - - !—b$

Initial length of the interval $L_0 = b - a$

$L_n = \frac{b-a}{F_n} = \frac{L_0}{F_n}$implies that

$\frac{L_n}{L_0} = \frac{1}{F_n}$

Fibonacci Method

1. Determine the number of iterations.
2. Define $L_2^* = \frac{F_{n-2}}{F_n} L_0 \; OR \; \frac{F_{n-1}}{F_n} L_0$, where $L_0 = b - a$ is the length of the initial interval.

Choose the two initial points x_1 and x_2 be such that

$$x_1 = a + L_2^* = a + \frac{F_{n-2}}{F_n} L_0 \quad (1)$$

$$x_2 = b - L_2^* = b - \frac{F_{n-2}}{F_n} L_0 \quad (2)$$

$a - - - - x_1 - - - - - - - - - - - - - - - - x_2 - - - - b$

Therefore,

$b - x_2 = x_1 - a$

If adding (1) and (2) , we have

Implies that

$b - x_2 = x_1 - a$

Which we get x_1 and x_2 and then we compute $f(x_1)$ and $f(x_2)$ and apply unimodal principal.

Unimodal principal

Let the initial interval [a,b] be given and the number of iteration be n

(i) Set $L = a$;$R = b$;$k = 1$ (iteration number)

(ii) Compute x_1 and x_2

If $f(x_2) > f(x_1)$ then $L_{k+1} = L_{k\,;}$; $R_k = x_2$ then

$a - - - x_1 - - - - - - - - x_2 - - - b$

$\leftarrow - - \times - - \rightarrow$

Therefore, new interval is (a,x_1)

If $f(x_2) < f(x_1)$ then $L_{k+1} = xL_{k\,;}$; $R_{k+1} = R_k$ then

$a - - - x_1 - - - - - - - - x_2 - - - b$

$\leftarrow - \times - \rightarrow$

Therefore, new interval is (x_1, b)

Example1. Minimize $f(x) = x^2$ over $[-5,15]$ using Fibonacci search method. Take $n = 7$.

Solution. Minimize $f(x) = x^2$over $[-5,15]$, $n = 7$.

Fibonacci numbers are

$F_0, F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8$……..

1 , 1 , 2 , 3 , 5 , 8 , 13 , 21 , 34 , …….

$-5 - - - - x_1 - - - - - - - - - - - - - x_2 - - - - 15$

K	$\frac{F_{n-k}}{F_{n-k+1}}$	L	R	x_1	x_2	$f(x_1)$	$f(x_2)$	L OR R	
1	$\frac{F_{7-1}}{F_{7-1+1}} = \frac{F_6}{F_7} = \frac{13}{21}$	-5	15	2.6191	7.3809	6.8596 Minimum	54.4772	L	
2	$\frac{F_{7-2}}{F_{7-2+1}} = \frac{F_5}{F_6} = \frac{8}{13}$	-5	7.3809	-5+7.3809 -2.6191=-0.2382	2.6191	0.567 Minimum	6.8596	L	
3	$\frac{F_{7-3}}{F_{7-3+1}} = \frac{4}{F_5} = \frac{5}{8}$	-5	2.6191	-2.1427	-0.2382	4.5911	0.0567 Minimum	R	
4	$\frac{F_{7-4}}{F_{7-4+1}} = \frac{F_3}{F_4} = \frac{3}{5}$	-2.1424	2.6191	-0.2382	0.7146	0.0567	0.5106	L	

5	$\frac{F_{7-5}}{F_{7-5+1}} = \frac{F_2}{F_3} = \frac{2}{3}$	-2.1427	0.7146	-1.1899	-0.2382	1.4158	0.0567	R	
6	$\frac{F_2}{F_3} = \frac{1}{2}$	-1.1899	0.7146	-0.2382	-0.2377	0.0567	0.0565	R	
7	$\frac{F_1}{F_2} = \frac{1}{1} = 1$	-0.2382	0.7146	-0.2377	0.7141	0.0565	0.5		

Therefore, Minimum $\in (-0.2382, 0.7141)$

$x^* = \frac{-0.2382+0.7141}{2} = 0.08945$

Calculations:

Now to find

$$x_2 = L + \left(\frac{F_{n-k}}{F_{n-k+1}}\right)(R-L) = -5 + \left(\frac{F_{7-1}}{F_{7-1+1}}\right)[15-(-5)] = -5 + \left(\frac{F_6}{F_7}\right)(20)$$

$$= -5 + \left(\frac{13}{21}\right)(20) = 7.38095$$

Therefore,

$$x_1 = L + R - x_2 = -5 + 15 - 7.38095 = 2.6191$$

Now

$$f(x_1) = f(2.06191) = (2.6191)^2 = 6.8596$$

$$f(x_2) = f(7.3809) = (7.3809)^2 = 54.4776$$

So $f(x_1)$ is minimum therefore, for next interval takink numbers on both sides of x_1.

Therefore, skipping R and so new interval is $(-5, 7.3809)$

$$\overset{-5}{L} - - - - x_1 - - - - - - - - - - - - - - - x_2 - - - - \overset{7.3809}{R}$$

And so on

Example2. Minimize $f(x) = x(x - 1.5)$ in $[0, 1]$ within the interval of uncertainty 0.25 of the initial interval of uncertainty.

Solution. Here in the question we have not given n. So to find n

$L_n \leq 0.25L_0$, implies that $\frac{L_n}{L_0} \leq 0.25$

But we know that $\frac{L_n}{L_0} = \frac{1}{F_n}$ which implies that $\frac{1}{F_n} \leq 0.25$OR $F_n \geq \frac{1}{0.25} OR\ 4$.

Therefore, taking smallest so $n = 4$.

Fibonacci numbers are

$F_0, F_1, F_2, F_3,\ F_4, F_5, F_6,\ F_7,\ F_8$……..

1 , 1 , 2 , 3 , 5 , 8 , 13 , 21 , 34 , ……..

$0 - - - - x_1 - - - - - - - - - - - - - x_2 - - - - 1.5$

Now to find for $k = 1$

$x_2 = L + \left(\frac{F_{n-k}}{F_{n-k+1}}\right)(R - L) = 0 + \left(\frac{F_{4-1}}{F_{4-1+1}}\right)[1 - 0)] = 0 + \left(\frac{F_3}{F_4}\right)(20)$

$= 0 + \left(\frac{3}{5}\right)(1) = 0.6$

Therefore,

$x_1 = L + R - x_2 = 0 + 1 - 0.6 = 0.4$

Now

$f(x_1) = f(0.4) = 0.4(0.4 - 1.5) = -0.44$

$f(x_2) = f(0.6) = 0.6(0.6 - 1.5) = -0.54$

So $f(x_2)$ is minimum therefore, for next interval taking numbers on both sides of x_2.

Therefore, skipping L and so new interval is $(0.4,\ 1)$

$$\underset{L}{0.4}- - - x_1 - - - - - - - - - - - - - - x_2 - - - - \underset{R}{1}$$

Now to find for $k = 2$

$$x_2 = L + \left(\frac{F_{n-k}}{F_{n-k+1}}\right)(R - L) = 0.4 + \left(\frac{F_{4-2}}{F_{4-2+1}}\right)[1 - 0.4)] = 0.4 + \left(\frac{F_2}{F_3}\right)(0.6) = 0.8$$

$$x_1 = L + R - x_2 = 0.4 + 1 - 0.8 = 0.6$$

And so on

K	$\frac{F_{n-k}}{F_{n-k+1}}$	L	R	x_1	x_2	$f(x_1)$	$f(x_2)$	L OR R	
1	$\frac{F_{4-1}}{F_{4-1+1}} = \frac{F_3}{F_4} = \frac{3}{5}$	0	1	0.4	0.6	-0.44	-0.54	R	
2	$\frac{F_{4-2}}{F_{4-2+1}} = \frac{F_2}{F_3} = \frac{2}{3}$	0.4	1	0.6	0.8	-0.54	0.56	R	
3	$\frac{F_{4-3}}{F_{4-3+1}} = \frac{F_1}{F_2} = \frac{1}{2}$	0.6	1	0.8	0.8	-0.56	-0.56	L	
4	$\frac{F_{4-4}}{F_{4-4+1}} = \frac{F_0}{F_1}$	0.6	0.8	0.8	0.6	-0.56	-0.56		

	$=\frac{1}{1}=1$								

Calculations:

Now to find

$$x_2 = L + \left(\frac{F_{n-k}}{F_{n-k+1}}\right)(R-L) = 0 + \left(\frac{F_{4-1}}{F_{4-1+1}}\right)[1-0)] = 0 + \left(\frac{F_3}{F_4}\right)(20)$$

$$= 0 + \left(\frac{3}{5}\right)(1) = 0.6$$

Therefore,

$$x_1 = L + R - x_2 = 0 + 1 - 0.6 = 0.4$$

Now

$$f(x_1) = f(0.4) = 0.4(0.4 - 1.5) = -0.44$$

$$f(x_2) = f(0.6) = 0.6(0.6 - 1.5) = -0.54$$

So $f(x_2)$ is minimum therefore, for next interval taking numbers on both sides of x_2.

Therefore, skipping L and so new interval is $(0.4,\ 1)$

$$\overset{0.4}{L} - - - x_1 - - - - - - - - - - - - - - x_2 - - - - \overset{1}{R}$$

And so on

$x_{min} \in [0.6, 0.8]$

So $x^* = \frac{0.6+0.8}{2} = 0.7$

$$f(x^*) = 0.7(0.7 - 1.5) = -0.56.$$

Example3. Perform 6 iterations of the Fibonacci search method to $Minimize\ f(x) = Maximize\left\{x^2, \frac{1-x}{2}\right\}$ over $[-1,1]$.

Solution. $Minimize\ f(x) = Maximize\left\{x^2, \frac{1-x}{2}\right\}$ means

$$Minimize\ f(x) = \begin{cases} \frac{1-x}{2} ; x < \frac{1}{2} \\ x^2 \quad ; \ x \geq \frac{1}{2} \end{cases}$$

$n = 6$

k	$\frac{F_{n-k}}{F_{n-k+1}}$	L	R	x_1	x_2	$f(x_1)$	$f(x_2)$	L OR R
1	$\frac{F_{6-1}}{F_{6-1+1}} = \frac{F_5}{F_6} = \frac{8}{13}$	-1	1	-0.2307	0.2307	0.6153	0.3846	R
2	$\frac{F_{6-2}}{F_{6-2+1}} = \frac{F_4}{F_5} = \frac{5}{8}$	-0.2307	1	0.2307	0.5386	0.3856	0.2901	R
3	$\frac{F_{6-3}}{F_{6-3+1}} = \frac{F_3}{F_4} = \frac{3}{5}$	0.2307	1	0.5386	0.6921	0.5386	0.4790	L
4	$\frac{F_{6-4}}{F_{6-4+1}} = \frac{F_2}{F_3}$	0.2307	0.6921	0.3842	0.5386	0.5377	0.2901	R

	$=\frac{2}{3}$							
5	$\frac{1}{2}$	0.3842	0.6921	0.5386	0.5377	0.2901	0.2831	L
6	$\frac{1}{1}=1$	0.3842	0.5377					

$Minimum \in (0.3842, 0.5377)$

$x^* = \frac{0.3842+0.5377}{2} = 0.46095$, therefore $f(x^*) = \frac{1-0.46095}{2} = 0.26953$

Golden Section Method

To find $x_2 = L + 0.618(R - L)$ and $x_1 = R + L - x_2$

Example1. Using Golden section method, minimize $f(x) = x^2$ over $[-5,15]$, taking $n = 7$.

Solution: Fibonacci numbers are

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, …….

$F_0, F_1, F_2, F_3, F_4, F_5, F_6, F_7, F_8, F_9$ ……..

$-5 - - - - x_1 - - - - - - - - x_2 - - - - 15$

Now

$x_2 = L + 0.618(R - L) = -5 + 0.618[15 - (-5)] = 7.36$ so

$x_1 = L + R - x_2 = -5 + 15 - 7.36 = 2.64$

k	L	R	x_1	x_2	$f(x_1)$	$F(x_2)$	Preserve L or R
1	-5	15	2.64	7.36	6.969 (Min.)	54.17	L
2	-5	7.36	-0.28	2.64	0.0784 (Min.)	6.9696	L
3	-5	2.64	-2.08	-0.28	4.3264	0.0784 (Min.)	R
4	-2.08	2.64	-0.28	0.84	0.0784	0.7056	L
5	-2.08	0.84	-0.96	-0.28	0.9216	0.0784	R
6	-0.96	0.84	-0.28	0.16	0.0784	0.0256	R
7	-0.28	0.84	0.16	0.4	0.0256	0.16	

Therefore,

$x_{min} \in [-0.28, 0.84]$

$x^* = \frac{-0.28+0.84}{2} = 0.28$

Minimum value $= (0.28)^2 = 0.0784$

Example2. Using golden section method, Minimize $f(x) = x(x-1.5)$ in $[0,1]$ with interval of uncertainty as 0.3.

Solution: Given that $\epsilon = 0.3$

$x_2 = L + 0.618(R-L)$ *and* $x_1 = R + L - x_2$

k	L	R	x_1	x_2	$f(x_1)$	$F(x_2)$	Preserve L or R
1	0	1	0.382	0.618	−0.42708	−0.54508 Min	R
2	0.382	1	0.618	0.764	−0.54508	−0.5623 Min	R
3	0.618	1	0.764	0.854	−0.5623 (Min.)	−0.55168	L
4	0.618	0.854	0.708	0.764	−0.56074	−0.5623	L

Thus $x_{min} \in [0.618, 0.854]$ and $x^* = \frac{0.618+0.854}{2} = 0.736$

Therefore, $f_{min} = 0.736(0.736-1) = -0.1943$

Example3. Using golden section method, Minimize $f(x) = Maximize\left\{x^2, \frac{1-x}{2}\right\}$ over $[-1,1]$ up to I_6.

Solution: Minimize $f(x) = Maximize\left\{x^2, \frac{1-x}{2}\right\}$, $n = 6$ means

$$Minimize\ f(x) = \begin{cases} \frac{1-x}{2}, & x \le \frac{1}{2} \\ x^2, & x \ge \frac{1}{2} \end{cases}, n = 6$$

$x_2 = L + 0.618(R-L)$ *and* $x_1 = R + L - x_2$

k	L	R	x_1	x_2	$f(x_1)$	$F(x_2)$	Preserve L or R
1	−1	1	−0.236	0.236	0.618	0.382	R

2	-0.236	1	0.236	0.528	0.382	0.27878	R
3	0.236	1	0.528	0.708	0.27878	0.50126	L
4	0.236	0.708	0.416	0.528	0.292	0.27878	R
5	0.416	0.708	0.528	0.596	0.27878	0.35522	L
6	0.416	0.596	0.484	0.528	0.258	0.27878	

Therefore, $x_{min} \in [0.416, 0.484]$ so $x^* = \frac{0.416+0.596}{2} = 0.506$

Thus, Minimum value of $f(x) = (0.506)^2 = 0.256036$

Newton's Method for minimization of functions for single variable

In the problem of minimizing a function $f(x)$ of a single variable x. Assuming that each measurement point x_k we can calculate $f(x_k)$, $f^{/}(x_k)$ and $f^{\cdot\cdot\cdot}(x_k)$.

We can fit a quadratic function through x_k that matches its first and second derivatives with that of the function (x) .

$q(x) = f(x_k) + (x - x_k) f^{!}(x_k) + \frac{1}{2}(x - x_k)^2 f^{!!}(x_k)$ such that

$q(x_k) = f(x_k)$, $q^{!}(x_k) = f^{!}(x_k)$ $and q^{!!}(x_k) = f^{!!}(x_k)$

Instead of minimizing $f(x)$, we minimize its approximation $q(x)$.

The first order necessary condition for a minimize of $q(x)$ yields

$q^{!}(x) = 0$, implies that

$f^{!}(x_k) + (x - x_k) f^{!!}(x_k) = 0$

In this taking $x = x_{k+1}$, we get

$$\boldsymbol{x_{k+1} = x_k - \frac{f^{!}(x_k)}{f^{!!}(x_k)}, \; k = 0, 1, 2, 3, \ldots..}$$

Example1. Using Newton's method, find the minimize of $f(x) = \frac{1}{2}x^2 - sinx$. The initial value is $x^{(0)} = \frac{1}{2}$. The required accuracy is $\epsilon = 10^{-5}$.

(In the sense that we stop when $|x_{k+1} - x_k| < \epsilon$.

Solution: Given function is

$$f(x) = \frac{1}{2}x^2 - sinx \quad (1)$$

Differentiating this, we get

$$f^{!}(x) = x - cosx \quad (2)$$

and

$$f^{!!}(x) = 1 + sinx \quad (3)$$

Given initial point $x^{(0)} = \frac{1}{2} = 0.5$

Therefore, from recurrence formula on taking $k = 0$, we get

$$x_1 = x_0 - \frac{f^{!}(x_0)}{f^{!!}(x_0)} = 0.5 - \frac{0.5 - \cos(0.5)}{1 + \sin(0.5)} = 0.5 - \frac{-0.3775}{1.479} = 0.7552$$

Proceeding in a similar way, we get

$$x_2 = x_1 - \frac{f^{!}(x_1)}{f^{!!}(x_1)} = 0.7552 - \frac{0.7552 - \cos(0.7552)}{1 + \sin(0.7552)} = 0.7552 - \frac{0.0271}{1.685} = 0.7391$$

$$x_3 = x_2 - \frac{f^{!}(x_2)}{f^{!!}(x_2)} = 0.7391 - \frac{0.7391 - \cos(0.7391)}{1 + \sin(0.7391)} = 0.7391 - \frac{9.461 \times 10^{-5}}{1.673} = 0.7390$$

$$x_4 = x_3 - \frac{f^{!}(x_3)}{f^{!!}(x_3)} = 0.7390 - \frac{0.7390 - \cos(0.7390)}{1 + \sin(0.7390)} = 0.7390 - \frac{1.17 \times 10^{-9}}{1.673} = 0.7390$$

Now

$$|x_4 - x_3| < \epsilon = 10^{-5}$$

$$f^{!}(x_4) = -8.6 \times 10^{-6} \approx 0,\ f^{!}(x_4) = 1.763 > 0$$

So x_4 is a point of minima

Minimum value is $f(x_4) = \frac{1}{2}(x_4)^2 - sin(x_4)$

Example2. Find the minimum value of $f(x) = x^3 - 3x - 7$, using Newton's method taking initial approximation $x_1 = 0.5\ and\ \epsilon = 0.001$.

Solution: Given function is $f(x) = x^3 - 3x - 7$ and initial approximation is $x_1 = 0.5$

For Newton's method recurrence formula is

$$x_{k+1} = x_k - \frac{f^{!}(x_k)}{f^{!!}(x_k)},\ k = 1, 2, 3, \ldots.$$

Now

$$f(x) = x^3 - 3x - 7$$

$$f^{!}(x) = 3x^2 - 3$$

$$f^{!!}(x) = 6x$$

Taking $k = 1$,we get

$$x_2 = x_1 - \frac{f^{!}(x_1)}{f^{!!}(x_1)} = x_1 - \frac{3(x_1)^2 - 3}{6x_1}$$

$$x_2 = 0.5 - \frac{3\times(0.5)^2 - 3}{6\times 0.5} = 0.5 - \frac{-2.25}{3} = 0.5 + 0.75 = 1.25$$

$$x_3 = x_2 - \frac{f^{!}(x_2)}{f^{!!}(x_2)} = x_2 - \frac{3(x_2)^2 - 3}{6x_2}$$

$$x_3 = 1.25 - \frac{3\times(1.25)^2 - 3}{6\times(1.25)} = 1.25 - \frac{1.6875}{7.5} = 1.475$$

$$x_4 = x_3 - \frac{f^{!}(x_3)}{f^{!!}(x_3)} = 1.475 - \frac{3\times(1.475)^2 - 3}{6\times(1.475)} = 1.475 - 0.39851412$$

$$x_4 = 1.076486$$

$$x_5 = x_4 - \frac{f^{!}(x_4)}{f^{!!}(x_4)} = x_4 - \frac{3(x_4)^2 - 3}{6x_4} = 1.076486 - \frac{3(1.076486)^2 - 3}{6\times 1.076486}$$

$$x_5 = 1.002716$$

$$x_6 = x_5 - \frac{f^{!}(x_5)}{f^{!!}(x_5)} = 1.0027 - \frac{3\times(1.0027)^2 - 3}{6\times(1.0027)} = 1.0027 - 0.002691$$

$$x_6 = 1.000009$$

$$x_7 = x_6 - \frac{f^{!}(x_6)}{f^{!!}(x_6)} = x_6 - \frac{3(x_6)^2 - 3}{6x_6} = 1.000009 - \frac{3(1.000009)^2 - 3}{6\times 1.000009}$$

$$x_7 = 1.000009 - 0.000009 = 1$$

Now

$|x_7 - x_6| = 0.000009 < 0.001$

$f^{!!}(x_7) = 6x_7 = 6 \times 1 = 6 > 0$

So x_7 is a point of minima.

Therefore point of minima is $x^* = x_7 = 1$

Minimum value is $= (1)^3 - 3 \times (1) - 7 = -9$.

Minimization of Multivariable Functions

Steepest Descent Method

Example 1. Minimize $f(x_1,x_2) = 2(x_1)^2 + 2x_1x_2 + (x_2)^2 + x_1 - x_2$. Consider the starting point as $(0\ 0)$,

Solution. Given function is

$f(x_1,x_2) = 2(x_1)^2 + 2x_1x_2 + (x_2)^2 + x_1 - x_2$ and initial starting point is $x_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

Grad f OR $\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 4x_1 + 2x_2 + 1 \\ 2x_1 + 2x_2 - 1 \end{pmatrix}$

$[\nabla f]_{X=X_0} = \begin{pmatrix} 0+0+1 \\ 0+0-1 \end{pmatrix} = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$

Therefore,

$s_0 = -[\nabla f]_{X=X_0} = -\begin{pmatrix} 1 \\ -1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$

Now next approximation

$$X_1 = X_0 + \lambda_0 s_0 \text{where} \lambda_0 = \frac{(s_0)^T s_0}{(s_0)^T A\, s_0} \text{ and } A = \begin{pmatrix} \frac{\partial^2 f}{\partial x_1{}^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2{}^2} \end{pmatrix} = \begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}$$

Therefore,

$$\lambda_0 = \frac{(-1 \quad 1)\begin{pmatrix} -1 \\ 1 \end{pmatrix}}{(-1 \quad 1)\begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} -1 \\ 1 \end{pmatrix}} = \frac{2}{2} = 1$$

So

$$X_1 = X_0 + \lambda_0 s_0 = \begin{pmatrix} 0 \\ 0 \end{pmatrix} + 1\begin{pmatrix} -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}$$

To check the optimality, we check

$$[\nabla f]_{X=X_1} = \begin{pmatrix} 4x_1 + 2x_2 + 1 \\ 2x_1 + 2x_2 - 1 \end{pmatrix} = \begin{pmatrix} -4 + 2 + 1 \\ -2 + 2 - 1 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Therefore, we move further

Now we find next iteration

$$X_2 = X_1 + \lambda_1 s_1 \text{where} s_1 = -[\nabla f]_{X=X_1} = -\begin{pmatrix} 4x_1 + 2x_2 + 1 \\ 2x_1 + 2x_2 - 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\lambda_1 = \frac{(s_1)^T s_1}{(s_1)^T A\, s_1} = \frac{(1 \quad 1)\begin{pmatrix} 1 \\ 1 \end{pmatrix}}{(1 \quad 1)\begin{pmatrix} 4 & 2 \\ 2 & 2 \end{pmatrix}\begin{pmatrix} 1 \\ 1 \end{pmatrix}} = \frac{1}{5}$$

So

$$X_2 = X_1 + \lambda_1 s_1 = \begin{pmatrix} -1 \\ 1 \end{pmatrix} + \frac{1}{5}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} -\frac{4}{5} \\ \frac{6}{5} \end{pmatrix}$$

To check the optimality, we check

$$[\nabla f]_{X=X_2} = \begin{pmatrix} 4x_1 + 2x_2 + 1 \\ 2x_1 + 2x_2 - 1 \end{pmatrix} = \begin{pmatrix} 4\left(-\frac{4}{5}\right) + 2\left(\frac{6}{5}\right) + 1 \\ 2\left(-\frac{4}{5}\right) + 2\left(\frac{6}{5}\right) - 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{5} \\ -\frac{1}{5} \end{pmatrix} \neq \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

Therefore this is not an optimal solution. We move further

$X_3 = X_2 + \lambda_2 s_2$and so on , we have

Iteration number	$(X_i)^T$	$(\nabla f)_{X=X_i}$	$(s_i)^T$	λ_i	$X_{i+1} = X_i + \lambda_i s_i$
0	$(0\ \ 0)$	$(1\ \ -1)$	$(-1\ \ 1)$	1	$(-1\ \ 1)$
1	$(-1\ \ 1)$	$(-1\ \ -1)$	$(1\ \ 1)$	0.2	$(-0.8\ \ 1.2)$
2	$(-0.8\ \ 1.2)$	$(0.2\ -0.2)$	$(-0.2\ 0.2)$	1	$(-1\ \ 1.4)$
3	$(-1\ \ 1.4)$	$(-0.2\ 0.2)$	$(0.2\ 0.2)$	0.2	$(0.961\ \ 1.44)$
4	$(0.96\ \ 1.44)$	$(1.72\ 3.8)$	$(-1.72\ \ -3.8)$	0.193	$(-0.59\ \ 0.73)$
5	$(-0.8\ \ 1.2)$	$(0.31\ -0.69)$	$(-0.31\ \ 0.69)$	1.24	$(-0.9344\ \ 1.407)$
6	$(-0.914\ \ 1.497)$	$(0.337\ \ 0.166)$	$(-0.337\ \ 0.166)$	0.192	$(-0.979\ \ 1.465)$
7	$(-0.979\ \ 1.465)$	$(0.01366\ -0.0277)$	$(-0.01366\ \ 0.0277)$	1.24	$(-0.999\ \ 1.4998)$
8	$(-0.996\ \ 1.4998$	$(0.0147\ 0.0077)$	$(-0.0147\ -0.007)$	0.192	$(-0.999\ \ 1.49)$

We can check where we have to stop and for this we have to check one of the following

(i) $\left|\frac{f(x_{i+1})-f(x_i)}{f(x_i)}\right| < \epsilon$

(ii) $\left\|\left|\nabla f\right|_{X=X_i}\right\| < \epsilon$

(iii) $\left|X_{i+1} - X_i\right| < \epsilon$

Minimum value at

$(-0.996\ \ 1.4998)$

So

Minimum value of $f(x_1,x_2) = 2(x_1)^2 + 2x_1x_2 + (x_2)^2 + x_1 - x_2$ is

$f(-0.996\ \ 1.4998) = 2(-0.996)^2 + 2(-0.996)(1.4998) + (1.4998)^2 + (-0.996) - (1.4998)$

$f(-0.996 \;\; 1.4998) = 2(0.992016) - 2.9876016 + 2.2494 - 0.996 - 1.4998$

$f(-0.996 \;\; 1.4998) = 4.2414 - 4.487409 = -0.2460$

Newtom's Method (Gradient Method)

Example 1. Minimize $f(x_1,x_2) = 2(x_1)^2 + 2x_1x_2 + (x_2)^2 + x_1 - x_2$ by taking initial approximation as $\begin{pmatrix}0\\0\end{pmatrix}$.

Solution. Recurrence relation for successive approximation is

$X_{i+1} = X_i - (H)^{-1}(\nabla f_I)$, where H is Hessian matrix and ∇f_I = grad $f = \begin{pmatrix}\frac{\partial f}{\partial x_1}\\ \frac{\partial f}{\partial x_2}\end{pmatrix}$ at X_i

And $H = \begin{pmatrix}\frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2}\\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2}\end{pmatrix}$

Initial approximation is $X_0 = \begin{pmatrix}0\\0\end{pmatrix}$, therefore

$(\nabla f)_{X=x_0}$ =grad $f = \begin{pmatrix}\frac{\partial f}{\partial x_1}\\ \frac{\partial f}{\partial x_2}\end{pmatrix} = \begin{pmatrix}4x_1 + 2x_2 + 1\\ 2x_1 + 2x_2 - 1\end{pmatrix}_{\begin{pmatrix}0\\0\end{pmatrix}} = \begin{pmatrix}1\\-1\end{pmatrix}$

$(H)_{X=X_0} = \begin{pmatrix}\frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2}\\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2}\end{pmatrix}_{X=X_0} = \begin{pmatrix}4 & 2\\ 2 & 2\end{pmatrix}$, therefore

$(H)^{-1} = \frac{1}{4}\begin{pmatrix}2 & -2\\ -2 & 4\end{pmatrix} = \begin{pmatrix}\frac{1}{2} & -\frac{1}{2}\\ -\frac{1}{2} & 1\end{pmatrix}$

Therefore

$X_1 = X_0 - (H)^{-1}(\nabla f) = \begin{pmatrix}0\\0\end{pmatrix} - \begin{pmatrix}\frac{1}{2} & -\frac{1}{2}\\ -\frac{1}{2} & 1\end{pmatrix}\begin{pmatrix}1\\-1\end{pmatrix} = \begin{pmatrix}-1\\ \frac{3}{2}\end{pmatrix}$

Now $(\nabla f)_{X=x_1}$ = grad $f = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \end{pmatrix} = \begin{pmatrix} 4x_1 + 2x_2 + 1 \\ 2x_1 + 2x_2 - 1 \end{pmatrix}_{\binom{0}{0}} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

If gradf= ∇f is = $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ then we stop.

If gradf= ∇f is not = $\begin{pmatrix} 0 \\ 0 \end{pmatrix}$ then we move to next iteration. $X_2 = X_1 - (H)^{-1}(\nabla f)$ and so on.

Direct Search Method (OR Non Gradient Method)

Simplex Method OR Nelder-Mead method

Algorithm

Centroid $\overrightarrow{X} = \frac{1}{n}\sum_{\substack{j=1 \\ j \neq r}}^{n+1} x_j$

Step 1 Reflection process

Calculate $X_r = \overrightarrow{X} + \alpha\left(\overrightarrow{X} - X_h\right)$, $\alpha > 0$

Step 2

After reflection if

$f(X_l) > f(X_r)$ →replacing X_h with X_r , otherwise go to step 4.

Step 3

If $f(X_l) > f(X_r)$ generating extended point

$X_e = \overrightarrow{X} + \gamma\left(X_r - \overrightarrow{X}\right)$, $\gamma > 1$ replacing X_h with X_e and go to step 2 for reflection.

Step 4

If $f(X_l) < f(X_r) < f(X_h)$ OR $f(X_r) < f(X_h)$

Find $X_{c_1} = \overrightarrow{X} + \beta\left(\overrightarrow{X} - X_w\right)$, $0 < \beta < 1$

$X_{c_0} = \overrightarrow{X} - \beta\left(\overrightarrow{X} - X_w\right)$, $0 < \beta < 1$

Check $f\left(X_{c_1}\right)$ & $f\left(X_{c_0}\right)$

$f(X_c) = Minimum\left\{f\left(X_{c_1}\right), f\left(X_{c_0}\right)\right\}$

Contraction

If the reflected point is worse then we try to contract towards a better solution;

Inside contraction

Outside contraction

Whichever is better.

Example 1. Using non-gradient method, minimize $(x_1,x_2) = 2(x_1)^2 + 2x_1x_2 + (x_2)^2 + x_1 - x_2$,

taking initial points as $X_1 = \begin{pmatrix}4\\4\end{pmatrix}, X_2 = \begin{pmatrix}5\\4\end{pmatrix}$, $X_3 = \begin{pmatrix}4\\5\end{pmatrix}$, $\alpha = 1$, $\beta = 0.5$, $\gamma = 2$. For convergence use the stopping criterion $\varepsilon = 0.2$.

Solution: Given function is

$f(x_1,x_2) = 2(x_1)^2 + 2x_1x_2 + (x_2)^2 + x_1 - x_2$

Therefore,

$f_1 = f(X_1) = 2(4)^2 + 2(4)(4) + (4)^2 + 4 - 4 = 80 \rightarrow$ Best (because lowest value)

$f_2 = f(X_2) = 2(5)^2 + 2(5)(4) + (4)^2 + 5 - 4 = 107 \rightarrow$ Worst (Because highest value)

$f_3 = f(X_3) = 2(4)^2 + 2(4)(5) + (5)^2 + 4 - 5 = 96 \rightarrow$ Good (Because between lowest and highest value)

Therefore,

$X_h = X_2$, $X_l = X_1$

Centroid $\overrightarrow{X} = \frac{1}{2}(X_1 + X_3) = \begin{pmatrix} 4 \\ 4.5 \end{pmatrix}$

Value of the function at $\overrightarrow{X}$ is

$f\left(\overrightarrow{X}\right) = 2(4)^2 + 2\,(4)\,(4.5) + (4.5)^2 + 4 - 4.5 = 87.75$

Now generating reflection point

$X_r = \overrightarrow{X} + \alpha\left(\overrightarrow{X} - X_h\right) = \begin{pmatrix} 4 \\ 4.5 \end{pmatrix} + 1.\left(\begin{pmatrix} 4 \\ 4.5 \end{pmatrix} - \begin{pmatrix} 5 \\ 4 \end{pmatrix}\right) = \begin{pmatrix} 3 \\ 5 \end{pmatrix}$

Value of the function at X_r is

$f\,(X_r) = 2(3)^2 + 2\,(3)\,(4.5) + (5)^2 + 3 - 5 = 71$

So $f\,(X_r) < f\,(X_l)\ OR\ f\,(X_1)$

[**Note**: – If $f\,(X_r) > f\,(X_l)\ OR\ f\,(X_1)$, the we find $X_{c_0} = \overrightarrow{X} - \beta\left(\overrightarrow{X} - X_h\right)$ and $X_{c_1} = \overrightarrow{X} + \beta\left(\overrightarrow{X} - X_h\right)$.
Then finding $f\left(X_{c_0}\right)$ and $f\left(X_{c_1}\right)$ and selecting the point for minimum of these two.]

Now, we find an extended point X_e

$X_e = \overrightarrow{X} + \gamma\left(X_r - \overrightarrow{X}\right) = \begin{pmatrix} 4 \\ 4.5 \end{pmatrix} + 2.\left(\begin{pmatrix} 3 \\ 5 \end{pmatrix} - \begin{pmatrix} 4 \\ 4.5 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 5.5 \end{pmatrix}$

Value of the function at X_e is

$f\,(X_e) = 2(2)^2 + 2\,(2)\,(5.5) + (5{,}5)^2 + 2 - 5.5 = 56.75$

Now

X_1 , $X_2\ OR\ X_e$, X_3

$X_1 = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$, $X_e\ (OR\ X_2) = \begin{pmatrix} 2 \\ 5.5 \end{pmatrix}$, $X_3 = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$

For convergence, we compute

Standard deviation $(\sigma) = \frac{1}{n}\sqrt{\sum_{i=1}^{3}\left(f(X_i) - f(\bar{X})\right)^2}$

$$= \frac{1}{3}\sqrt{\left[\left(f(X_1) - f(\bar{X})\right)^2 + \left(f(X_2) - f(\bar{X})\right)^2 + \left(f(X_3) - f(\bar{X})\right)^2\right]}$$

$$= \frac{1}{3}\sqrt{\left[(80 - 87.75)^2 + (56.75 - 87.75)^2 + (96 - 87.75)^2\right]} = 19.06$$

Standard deviation $\sigma = 19.06 > \varepsilon$

Therefore, moving to next iteration.

Iteration 2

Step-1

X_1 , X_2 OR X_e , X_3

$X_1 = \begin{pmatrix} 4 \\ 4 \end{pmatrix}$, $X_2 = \begin{pmatrix} 2 \\ 5.5 \end{pmatrix}$, $X_3 = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$

$f(X_1) = 80$, $f(X_2) = 56.75$, $f(X_3) = 96$

Therefore,

$X_w = X_3 = \begin{pmatrix} 4 \\ 5 \end{pmatrix}$ and $X_B = X_2 = \begin{pmatrix} 2 \\ 5.5 \end{pmatrix}$

Step 2

Centroid $\vec{X} = \frac{1}{2}(X_1 + X_2) = \begin{pmatrix} 3 \\ 4.75 \end{pmatrix}$

Value of the function at $\vec{X}$ is

$$f\left(\vec{X}\right) = 2(3)^2 + 2(3)(4.75) + (4.75)^2 + 3 - 4.75 = 67.31$$

Step 3

Now generating reflection point

$$X_r = \vec{X} + \alpha\left(\vec{X} - X_w\right) = \begin{pmatrix} 3 \\ 4.75 \end{pmatrix} + 1.\left(\begin{pmatrix} 3 \\ 4.75 \end{pmatrix} - \begin{pmatrix} 4 \\ 5 \end{pmatrix}\right) = \begin{pmatrix} 2 \\ 4.5 \end{pmatrix}$$

Value of the function at X_r is

$$f(X_r) = 2(2)^2 + 2(2)(4.5) + (4.5)^2 + 2 - 4.5 = 43.75$$

Step 4

So $f(X_r) < f(X_l)\ OR\ f(X_2)$ so we find expanded point X_e as

$$X_e = \vec{\overline{X}} + \gamma\left(X_r - \vec{\overline{X}}\right) = \begin{pmatrix} 3 \\ 4.75 \end{pmatrix} + 2.\left(\begin{pmatrix} 2 \\ 4.5 \end{pmatrix} - \begin{pmatrix} 3 \\ 4.75 \end{pmatrix}\right) = \begin{pmatrix} 1 \\ 4.25 \end{pmatrix}$$

Value of the function at X_e is

$$f(X_e) = 2(1)^2 + 2(1)(4.25) + (4.25)^2 + 1 - 4.25 = 25.3125$$

Step 5

As $f(X_e) < f(X_B)$, we replace X_e by X_3

Therefore,

$X_1\ , X_2\ , X_3\ OR\ X_e$

$$X_1 = \begin{pmatrix} 4 \\ 4 \end{pmatrix},\ X_2 = \begin{pmatrix} 2 \\ 5.5 \end{pmatrix},\ X_3 = \begin{pmatrix} 1 \\ 4.25 \end{pmatrix}$$

Step 6

For convergence, we compute

$$\text{Standard Deviation} = \frac{1}{n}\sqrt{\sum\left(f(X_i) - f(\overline{X})\right)^2}$$

$$= \frac{1}{3}\sqrt{\left[\left(f(X_1) - f(\overline{X})\right)^2 + \left(f(X_2) - f(\overline{X})\right)^2 + \left(f(X_3) - f(\overline{X})\right)^2\right]}$$

$$= \frac{1}{3}\sqrt{\left[(80 - 67.31)^2 + (56.75 - 67.31)^2 + (25.3125 - 67.31)^2\right]} = 26.1 > \varepsilon$$

Now we go to next iteration and so on.